

LLIFS Specification

Draft v1.0

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LLIFS Specification

Status: Draft (v1.0-draft). All sections (§§1-17) drafted, section-reviewed, reconciled end-to-end (terminology normalized, requirement IDs collision-free, encodings byte-exact), and revised against two external technical reviews: the first (memory byte model, proof-block taxonomy, delta self-containment, workload-identity breadth, yield semantics, local-CAS integrity, directfs, inline reserved in v1, the Phase-1 density-gate ladder); the second adding the per-agent runtime-state delta (DELTA-2R / RDELTA / LLRD1), virtual zero-subtree proof rules (SPARSE-8), startup-cohesion reserved in v1, an exact oci_image_digest form (ENC-BD-4), LLMD1 source-precedence (ENC-MD-5), a named compatibility verifier step (MATRIX-3), and the gVisor page-provider distinction (GVISOR-6). Phase 1 prototyping is green-lit from this draft. HARD FREEZE BLOCKERS (v1.0 MUST NOT freeze until done): add the runtime delta (done in this draft, pending implementation proof); independently reproduce the §15.3 Terrapin conformance vectors via a second oracle (ENC-CONF-2); and reconcile the §15.8 LLWI1 field set against ateleit.WorkloadSpec (ENC-WI-1) – a workload-identity mismatch is a restore-safety bug. System: LLIFS (a verified, deduped, lazily-faulted state substrate for high-density agent FaaS). Primary target: gVisor (runsc) checkpoint/restore; Firecracker microVM next. One content-addressed, Terrapin-verified store delivers two state planes: a filesystem plane and a memory plane. External identity: OCI-compatible sha256. Internal identity: terrapin-sha256, profile llifs-terrapin-sha256-v2. Dependency: Terrapin Specification v0.3 (the hashing primitive). Its profile parameters and the load-bearing conformance constants are inlined here (§2.2, §15); the algorithm itself is normative-by-reference.

Introduction

LLIFS is the storage and state-management layer for running very large numbers of sandboxed agent workloads on a single fleet. It exists to make two things simultaneously true: a sandbox starts in roughly the time it takes to fault in the few blocks it actually touches, and every byte the sandbox is ever shown has been cryptographically verified against a trusted identity before execution sees it. Fast start and verified state are not traded off against each other; the design delivers both at once.

The setting is agent FaaS: thousands to millions of agent instances that are near-identical, the same base image and the same warmed runtime (interpreter, framework, model client), differing only in a small amount of per-agent state. Treating each instance as an independent copy of a full filesystem and a full memory image does not scale. It is too slow to start, because you move whole images before the first useful instruction runs, and too expensive to keep, because you store the same bytes a million times. The core thesis follows from this: do not optimize "pull the whole state faster." Start the workload after fetching only the bytes required for useful execution, while verifying every exposed byte, and make the shared part genuinely shared so that a million near-identical agents collapse to one verified base plus a small per-agent delta.

The approach. LLIFS keeps exactly one verified, content-addressed copy of each distinct byte-range and composes every sandbox from a shared base plus a small private overlay:

- Two state planes. A filesystem plane (the rootfs the workload reads and writes) and a memory plane (a warmed guest-memory image a sandbox restores from). Both are built from one content-addressed store.
- Lazy, verified fault-in. Bytes are fetched and verified in fixed blocks on demand, so a workload begins as soon as the blocks on its startup path are present, not after a whole image transfers. Nothing reaches the workload until its containing block has verified (§8).
- Shared copy-on-write base. One base filesystem and one base memory snapshot back many sandboxes; only the files and pages an agent actually changes become private (§4.5, §5, §6).
- Content-addressed identity end to end. Every internal object is named by its content (§2), so identical content is automatically a single object; this is what collapses near-identical agents into one base plus deltas.
- A per-snapshot State Root. The complete, restorable identity of one agent snapshot (its base, its memory delta, its filesystem delta, its runtime delta, and the runtime it is pinned to) is itself a single content-addressed value with a parent pointer, giving a lineage DAG and cheap forking (§6).

Two identity planes. LLIFS speaks two digest "languages" and never confuses them (§2). The external plane is OCI-compatible sha256, so images, registries, scanners, signatures, and admission policy interoperate unchanged. The internal plane is terrapin-sha256 (profile llifs-terrapin-sha256-v2): the verified, length-bound, tree-structured identity over which all fetch, dedup, and copy-on-write decisions are made. Terrapin itself, the hashing primitive, is a separate dependency; this document inlines only its profile parameters and load-bearing constants (§2.2, §15) and references the algorithm normatively.

Primary target. The first runtime is gVisor (runsc) checkpoint/restore, with Firecracker microVMs as the next adapter (§11). The design reuses existing runtime restore machinery wherever it can; the one capability it needs that does not exist yet (a shared copy-on-write memory base) is stated as a runtime requirement (§5, §11).

What this document is. This is the single normative specification for LLIFS. It defines the object model and identity scheme (§2), the two state planes (§§3-5), the agent lifecycle (snapshot, reset, hibernate, yield, resume, and fork) (§§6-7), the distribution, caching, and garbage-collection model (§§8-10), the control-plane binding (§12), the security model (§14), and, so that independent implementations agree byte-for-byte, the canonical on-the-wire encodings and conformance vectors (§15). It consolidates a family of earlier design documents into one; all cross-references are internal section references.

What this document is not. It does not define the Terrapin hashing algorithm (a referenced dependency), a scheduler or registry implementation, or any deployment-specific policy. Items the operator must supply are called out as out of scope (§14.4). The requirement keywords are normative per §1.

How to read it. Read this introduction, then §1 (normative language) and §2 (identity and the canonical vocabulary). §2 and the glossary (§17) are the terminology anchor for the whole document: every other section uses those terms and no synonyms. From there, §§3-5 define the filesystem and memory planes; §§6-7 the agent lifecycle, State Root, fork, and resume; §§8-10 fetch, verification, caching, GC, and scheduling; §§11-13 runtime adapters, control-plane binding, and observability; §14 the security model; §15 the canonical encodings and conformance vectors (the conformance surface for an implementation); §16 the phasing and minimum viable surface; and §17 the glossary.

Document outline

1. Normative language
2. Identity and digest model [complete]
3. Materialization and the logical rootfs tree (LLT1) [complete]
4. Filesystem plane: content-addressed layout (LLAY1) [complete]
5. Memory plane: base snapshot, copy-on-write restore [complete]
6. Agent state and lifecycle: overlay, delta, State Root, lineage, fork, reset/hibernate/yield [complete]
7. Startup and resume: profiles, packs, bootstrap, TTFE/TTR [complete]
8. Fetch, verification, transport [complete]
9. Cache, cache domains, leases, garbage collection [complete]
10. Scheduler integration and warmth [complete]

11. Runtime adapters (gVisor, Firecracker)	[complete]
12. Control-plane binding	[complete]
13. Observability	[complete]
14. Security model	[complete]
15. Canonical encodings and conformance vectors	[complete]
16. Phase plan and minimal viable surface	[complete]
17. Glossary	[complete]

1. Normative language

The keywords **MUST**, **MUST NOT**, **REQUIRED**, **SHALL**, **SHALL NOT**, **SHOULD**, **SHOULD NOT**, **RECOMMENDED**, **MAY**, and **OPTIONAL** are to be interpreted as normative requirements.

2. Identity and digest model

LLIFS has two digest planes and one canonical object vocabulary. This section is the terminology anchor for the whole document: every other section uses the canonical terms defined in §2.3 and §17, and no synonyms.

2.1 Two digest planes

External digest plane: for compatibility with OCI registries, descriptors, image manifests/indexes, config and layer blobs, OCI artifact descriptors, scanners, SBOMs, signatures, and admission policy. The external digest algorithm is sha256.

Internal digest plane: for LLIFS-native identity and verification. The internal digest algorithm is terrapin-sha256, profile llifs-terrapin-sha256-v2. It is the identity of every internal object: content objects, base memory snapshots, base descriptors, deltas, State Roots, the logical rootfs tree and physical layout encodings, and the runtime-state blob (startup cohesion is reserved in v1, §4.4). (Hash-file/proof blocks are not separate objects: they are the level ≥ 1 blocks of an owning object, verified through its tree, TERRAPIN-6.) Local cache keys (§9) are DERIVED from these identifiers (directly, or HMAC-keyed per cache domain); they are not themselves objects.

There is exactly one internal identifier scheme: terrapin-sha256:<64 lowercase hex>. There is no separate per-object-class digest scheme; objects differ by their content and their declared class, not by their hash namespace.

DIGEST-1:

External OCI object identity **MUST** use sha256.

DIGEST-2:

Internal LLIFS object identity **MUST** use terrapin-sha256 (profile llifs-terrapin-sha256-v2). The bare recursive tree root **MUST NOT** be used as an identifier (it is not self-describing about length or tree height).

DIGEST-3:

External and internal identifiers **MUST NOT** be substituted for one another. The OCI digest proves registry object identity; the Terrapin digest proves LLIFS content identity (§2.5).

2.2 Terrapin profile (llifs-terrapin-sha256-v2)

Parameters (fixed for this profile):

algorithm: terrapin-sha256

```

block size: 2097152 bytes exactly (2 MiB; NOT 2,000,000)
leaf hash:  gitoid-sha256
node hash:  gitoid-sha256
identifier:  G(canonical Terrapin root manifest)
digest form: terrapin-sha256:<64 lowercase hex>

```

Git0ID SHA-256:

```
G(data) = sha256("blob " || decimal(len(data)) || "\0" || data)
```

where decimal(n) is base-10 ASCII with no leading zeros and "\0" is one NUL byte. G(data) is 32 bytes. The "blob <len>\0" framing binds input length into the digest, detecting truncation and short writes at every level.

The Terrapin identifier is G over a canonical four-field ASCII manifest (terrapi, block_size, length, tree), LF-terminated, in that fixed order. A manifest that does not parse to exactly the canonical form MUST be rejected, not normalized. (Full manifest grammar and accept/reject matrix: §15.)

Load-bearing constants (the rest of the vector set is in §15):

```

G(empty)                = 473a0f4c3be8a93681a267e3b1e9a7dcda1185436fe141f7749120a303721813
G(2 MiB zero block) [leaf] = 67cbcd9b97ddabde2863f4daefa4f57176567a7c3ccfa1560c1065f9c8af74d6
identifier("hello world", len 11) = terrapin-sha256:7bc0163f32e5f6082308ae0dff3dc7c9b0488e5aa652d9de01418df5ec800c8c

```

TERRAPIN-1:

New LLIFS objects MUST use profile llifs-terrapi-sha256-v2.

TERRAPIN-2:

imagefsd MUST verify, for every object, that G(canonical Terrapin manifest) equals the object's terrapi-sha256 identifier.

TERRAPIN-3:

The fixed 2 MiB block is the unit of fetch and verification (a "block", §2.3). No sub-block proofs and no sub-2-MiB profile are defined; finer granularity, if ever needed, is a separate profile justified by measured over-fetch, never by a desire for finer dedup (dedup is achieved structurally: §2.3, §4, §5).

TERRAPIN-4:

Bytes MUST NOT be exposed to a workload until the containing block has verified against a trusted Terrapin identifier. Peers, mirrors, registries, caches, and packs are transport, never trust anchors.

TERRAPIN-5:

A content object of size ≤ 2 MiB has no hash-file/proof blocks: the data block is the leaf and the tree root is G(data). Only multi-block objects carry level ≥ 1 hash-file blocks; one 2 MiB level-1 block authenticates up to a 128 GiB contiguous region.

TERRAPIN-6:

Hash-file/proof blocks are NOT independent objects: they are the level ≥ 1 blocks of an owning Terrapin object, addressed by BlockRef (object / level / blockIndex, §2.3 block) and verified through that object's manifest and tree. Only the object classes of §2.3 (and the structural plane objects of §2.3 intro) are Terrapin objects with their own identifier; data blocks (level 0) and proof blocks (level ≥ 1) are addressed under those objects and have no identifier of their own.

2.3 Object taxonomy (canonical vocabulary)

These are the canonical nouns for agent state, and the only terms this document uses for it. Two adjacent categories are deliberately NOT in this list and are not agent-state object classes: (a) STRUCTURAL plane objects: the logical rootfs tree encoding (§3), the physical layout encoding (§4), and the runtime-state blob and memory layout (§5), which are Terrapin objects, each defined in its own section; and (b) COMMITTED FIELDS: sandbox pin, workload identity, run mode, and producer, which are committed inside a descriptor or a State Root, not stored as standalone objects. Terrapin proof (hash-file) blocks are NOT objects: they are the level ≥ 1 blocks of an owning object, addressed by BlockRef and verified through that object's manifest/tree (§2.2 TERRAPIN-6), never independently identified. Local cache keys (§9) are derived, not objects. Synonyms used in earlier drafts (chunk, chunk map, artifact, ⇨ plane manifest, opaque checkpoint, golden memory as a distinct object) are retired; see the Glossary (§17) for the mapping.

block

A 2,097,152-byte Terrapin block (the final block of a level MAY be short): data at level 0, hash-file/proof at level 1+. The unit of fetch and verification into the node CAS. A block is addressed within an object as object-digest / level / blockIndex (a BlockRef); it has no independent identifier (TERRAPIN-6).

content object

A Terrapin object whose bytes are exactly the content of one logical regular file. Its identifier is terrapin-sha256 over the file's logical (hole-expanded) content bytes, independent of path, mode, owner, mtime, xattrs, or which image references it. The unit of filesystem dedup. Whole-file: sub-file content-defined chunking is NOT used (it is a reserved future layout). Two files with identical content resolve to one content object within a cache domain.

base rootfs

The read-only, deduped filesystem base. It is (a) a canonical logical rootfs tree: paths, types, modes, owners, mtimes, xattrs, symlink/device/FIFO nodes, hardlink groups, encoded as LLT1, yielding the logicalRootfsDigest (§3); and (b) that tree's regular-file content realized as content objects via the content-addressed physical layout LLAY1, yielding the layoutDigest (§4). Shared across agents within a cache domain.

base memory snapshot

The read-only guest-memory image of a warmed sandbox (e.g. an agent runtime that has loaded its interpreter, framework, and client and reached a restore point), stored as a single Terrapin object in guest-address-linear order (offset 0 in the object is guest offset 0). Identity is terrapin-sha256 over the canonical image bytes. Shared copy-on-write across agents (§5). "Golden" is an informal adjective for a base memory snapshot of a pre-warmed shared runtime; it is not a distinct object class.

base descriptor

The signed identity object binding one base, as one unit: its OCI image digest, base rootfs (logicalRootfsDigest + layoutDigest), base memory snapshot, the non-memory runtime-state blob needed to resume (threads, fds, namespaces, vCPU/sentry state), the memory layout, the sandbox pin (§2.5), the runtime compatibility matrix (§11.4), and the assurance mode. (The full committed field list and encoding are §2.5 DIGEST-BIND-3 and §15.5 LLBD1.) A base descriptor is builder- or derived-attested and is the shared, multi-agent base identity. The (rootfs + memory + runtime-state + layout) tuple MUST NOT be recombinaible across bases (it is committed as a unit).

overlay

A live, per-agent writable layer over a base while the agent runs: copy-on-write dirty guest-memory pages plus the writable filesystem upper. An overlay is ephemeral (RAM and node-local scratch); it MUST NOT be written to the shared CAS and receives no dedup or warmth credit while the agent is running.

delta

The persisted, content-addressed snapshot of an overlay, captured at hibernate or fork. It has three parts. The MEMORY DELTA is a page MAP: for each diverged guest page, the source of its bytes (this delta's data object, an ancestor's, the base, or the canonical zero) and range, so every page resolves by direct reference with no ancestor traversal (§6.3) and a fork shares parent/base pages zero-copy (§15 LLMD1). The FILESYSTEM DELTA is a canonical overlay-delta object that reconstructs the writable upper completely: content-object references for changed regular-file data, plus every overlay namespace and metadata operation: created/changed directories, symlinks, device and FIFO nodes, ownership/mode/mtime/xattr changes, renames, and deletions (whiteouts/tombstones). (Copy-up breaks hardlink identity, as in default OverlayFS; the delta records files by content, not hardlink groups: §15 LLFD1.) The RUNTIME DELTA is an opaque runtime-native object capturing the diverged non-memory runtime state required for exact process resume (DELTA-2R, §15 LLRD1). A delta is per-agent and private: it lives in per-cache-domain-keyed CAS and is not deduped across cache domains.

State Root

The content-addressed identity of one persisted agent snapshot. It is a Terrapin object whose canonical encoding (§15) binds: the actor identity; a base descriptor reference; a memory delta reference; a filesystem delta reference; a runtime delta reference; the sandbox pin; the workload identity; the run mode; the

producer; the production time; and a parent State Root reference (the lineage edge). A State Root is minted at suspend, $\hookrightarrow$ hibernate, or fork, never for a merely-running agent (whose live per-agent state is an ephemeral overlay). It is the unit of resume identity and of forking.

lineage

The parent-pointer DAG over State Roots. Because a State Root's encoding commits its parent, a State Root identifier transitively commits to its ancestry. A fork creates a child State Root whose parent is the source snapshot and whose memory, filesystem, and runtime deltas begin as copy-on-write references to the source State Root's delta components (§6).

Relationships (informative):

```

base descriptor (shared, builder-attested)
    ▲
    | a State Root references its base descriptor
    |
State Root (per-agent, runtime-attested) → parent State Root → ... (lineage)
    |
    ├── memory delta      (page-indexed)      per-agent, private
    ├── filesystem delta  (overlay-delta object) per-agent, private
    └── runtime delta     (opaque runtime-native) per-agent, private

running agent  = base (shared, copy-on-write) + overlay (ephemeral, not stored)
persisted agent = State Root = base descriptor ref + delta (memory + filesystem +
                                     runtime) + lineage
  
```

OBJ-1:

Every stored LLIFS object, an agent-state object (above) or a structural plane object (§3, §4, §5), MUST be a Terrapin object identified per §2.2. Hash-file/proof blocks are not objects (they are the level ≥ 1 blocks of an owning object, addressed by BlockRef, §2.2 TERRAPIN-6); committed fields (§2.3 intro) are not separate objects.

OBJ-2:

A content object's identity MUST be over hole-expanded logical content bytes only; path, metadata, and hole structure MUST NOT affect it (§4).

OBJ-3:

A base memory snapshot MUST be stored in guest-address-linear order; pages MUST NOT be permuted in the stored object (addresses are load-bearing). Restore ordering is a fetch schedule, never a permutation of the object (§5).

OBJ-4:

A running agent's live per-agent state MUST be an overlay (ephemeral, not in the shared CAS). Persisted per-agent state MUST be a delta referenced by a State Root.

OBJ-5:

A State Root MUST commit actor, base descriptor, memory delta, filesystem delta, runtime delta, sandbox pin, workload identity, run mode, producer, created, and parent; absent components MUST be the explicit "none" value, never omitted (§15).

OBJ-6:

Identical objects (content objects, memory-delta, filesystem-delta, and runtime-delta components, base objects) MUST dedupe to one CAS entry per cache domain (§9).

2.4 Two granularities (the density principle)

LLIFS deliberately decouples two units that earlier designs conflated:

block (2 MiB): the unit of fetch and verification into the node CAS.
 page (4 KiB): the platform MMU page; the unit of per-agent copy-on-write sharing for the memory plane.

Dedup is achieved structurally, not by hashing ever-finer pieces:

- filesystem plane: by the content object (whole-file identity, §4);
- memory plane: by copy-on-write page sharing of one verified base across agents (§5), not by content-addressing each agent's pages.

This is why the memory base uses the ordinary 2 MiB profile yet still shares at

4 KiB: once a 2 MiB base block is verified-resident, its pages are shared copy-on-write across sandboxes with no re-fetch and no re-verification.

2.5 Binding chain

Internal Terrapin identities MUST be bound to a trusted root. There are two binding regimes, and they MUST NOT be substituted.

Builder-attested regime (the shared base). The signed BASE DESCRIPTOR is the attested object that ties the whole base back to the OCI trust root; it is the single signature subject:

```
OCI sha256 image digest
  → deterministic materialization (§3) → logicalRootfsDigest
    → layoutDigest (a physical layout implementing it, §4)
      → content objects
base memory snapshot + runtime-state blob + memory layout + sandbox pin (§5)
└─ the signed base descriptor commits ALL of the above as one unit (
    OCI digest, logicalRootfsDigest, layoutDigest, base memory snapshot,
    runtime-state blob, memory layout, sandbox pin, compatibility matrix, and
    assurance mode) so none can be recombined and the base memory snapshot
    (not derivable from the rootfs digests) is bound to the same OCI subject.
```

Runtime-attested regime (per-agent state). A memory delta, filesystem delta, runtime delta, and State Root are produced at runtime; they have no OCI subject and no builder.

Their integrity is unchanged (Terrapin all the way down); their provenance anchor is a runtime ATTESTATION ENVELOPE:

- signer: the producing node OR the control plane, each holding a key issued by a configured trust root within the node trust boundary. The signer MUST equal the producer recorded in the State Root, or be authorized by policy to attest on that producer's behalf;
- subject: the State Root identifier (which, being content-addressed, transitively covers ALL committed State Root fields: actor, parent, base-descriptor reference, all three delta references (memory, filesystem, runtime), sandbox pin, workload identity, run mode, producer, and created);
- validation: a verifier MUST check the envelope signature against a configured runtime trust root before exposing any runtime-produced byte, AND MUST independently validate the referenced base descriptor under the builder-attested regime (a trusted State Root does NOT confer trust on its base; both checks are required).

Sandbox pin: the runtime assets that produced a snapshot (e.g. the runsc binary) are content-addressed in the EXTERNAL plane by sha256 (they are ordinary fetched assets), recorded per architecture. The pin is a committed field of the base descriptor and of any State Root whose memory is restorable, so a restore on any node selects a compatible runtime and the scheduler can constrain placement (§5, §11). Its canonical encoding is defined in §15.

DIGEST-BIND-1:

A trusted materialization binding MUST bind an OCI sha256 image digest to a logicalRootfsDigest.

DIGEST-BIND-2:

A physical layout MUST declare the logicalRootfsDigest it implements; a verifier MUST reject a layout whose declared logical digest does not match.

DIGEST-BIND-3:

The signed base descriptor MUST be the single attested object binding, as one unit, the OCI image digest, logicalRootfsDigest, layoutDigest, base memory snapshot, runtime-state blob, memory layout, sandbox pin, compatibility matrix (§11.4), and assurance mode.

These MUST NOT be recombining across base descriptors, and the base memory snapshot MUST be bound to the same OCI subject through this descriptor (it is not derivable from the rootfs digests).

DIGEST-BIND-4:

A State Root, and the deltas it references, are runtime-attested by the attestation envelope (§2.5): a verifier MUST validate the envelope against a

configured runtime trust root before exposing any runtime-produced byte.

Builder-assurance modes MUST NOT be claimed for runtime-produced state.

DIGEST-BIND-5:

The builder-attested base and the runtime-attested per-agent state are distinct regimes and MUST NOT be substituted for one another.

DIGEST-BIND-6:

Every byte exposed to a workload MUST be verified against a Terrapin identifier reachable from a trusted, validated root: a base descriptor validated under the builder regime, or a State Root validated under the runtime regime. A trusted State Root does NOT confer trust on its referenced base descriptor; restore MUST validate the base descriptor independently (DIGEST-BIND-3) before exposing any base byte.

DIGEST-BIND-7:

A State Root whose memory is restorable, and the base descriptor it references, MUST commit a per-architecture sandbox pin; restore MUST reject an incompatible runtime and placement MUST honor it.

3. Materialization and the logical rootfs tree (LLT1)

The base rootfs (§2.3) has two identities: a LOGICAL identity (the filesystem tree, independent of storage) and a PHYSICAL identity (how it is stored as content objects, §4). This section defines the logical identity and the deterministic function that produces it from an OCI image.

3.1 OCI rootfs materialization

The materialization function maps an OCI image to a canonical logical rootfs tree. Identifier: `llifs-oci-rootfs-materialization-v1`.

Inputs: OCI image manifest digest and bytes; config descriptor and bytes; ordered layer descriptors; compressed layer blobs; uncompressed `diff_ids`; target platform. Output: the canonical logical rootfs tree (encoded per §3.2) and its `logicalRootfsDigest`.

- MAT-1: The target platform MUST be resolved before materialization.
- MAT-2: Layers MUST be applied in OCI manifest order for the selected platform.
- MAT-3: Each external OCI descriptor MUST be verified in the external plane (sha256).
- MAT-4: Each compressed layer MUST decompress to bytes whose digest matches the corresponding config `diff_id`; a mismatch MUST reject materialization.
- MAT-5: Tar entries MUST be interpreted as byte paths; implementations MUST NOT perform Unicode normalization.
- MAT-6: Absolute paths, paths containing a NUL byte, and paths containing ANY `".."` component MUST be rejected. LLIFS does not resolve `".."`: rather than canonicalize `"a/../b"`, any `".."` component rejects. This removes normalization ambiguity and rejects root escape with it.
- MAT-7: Path normalization MUST, component-wise: (a) drop `".."` components; (b) collapse repeated `"/"` to a single `"/"`; (c) strip any trailing `"/"`. It MUST NOT change the byte identity of valid components. The result is relative to root (no leading `"/"`, no trailing `"/"`, single `"/"` separators), and directory and file entries normalize identically.
- MAT-8: Within and across layers, a later entry for the same normalized path replaces an earlier one, subject to whiteout/opaque semantics.
- MAT-9: Whiteouts apply OCI semantics: `".wh.<name>"` removes `<name>` from the lower-visible tree; `".wh..wh..opq"` makes the containing directory opaque.
- MAT-10: Whiteout markers MUST NOT appear as visible entries.
- MAT-11: Opaque-directory markers MUST NOT appear as visible entries.
- MAT-12: Regular files, directories, symlinks, hardlinks, device nodes, FIFOs, ownership, mode bits, xattrs, mtimes, and link relationships MUST be represented when runtime-visible.
- MAT-13: Symlink targets MUST be stored as raw link bytes, unresolved.
- MAT-14: A hardlink whose target is absent or unsupported MUST reject.
- MAT-15: Hardlinks MUST be represented as the same content object plus a shared hardlink group, never as duplicated content.

- MAT-16: Unsupported tar/PAX/filesystem metadata that affects runtime-visible semantics MUST reject, not silently drop.
- MAT-17: Sparse files MUST be represented deterministically (§4 SPARSE).
- MAT-18: Device-node major/minor MUST be represented deterministically.
- MAT-19: logicalRootfsDigest MUST be independent of physical packing, fetch ordering, registry compression, and cache state.
- MAT-20: Materialization MUST be deterministic: given the same OCI image digest, platform, and function version, all conforming implementations MUST produce the same logicalRootfsDigest or reject. The byte-exact encoding of §3.2 is what makes this portable across implementations.

Assurance modes (how OCI→rootfs equivalence is established; a committed field of the base descriptor, §2.5):

asserted	A signer asserts OCI → logicalRootfsDigest → layoutDigest. The node verifies signatures and Terrapin bytes but does not re-derive the rootfs. Fastest, weakest; MUST be reported as a weaker mode and MUST NOT claim node-verified equivalence.
derived-attested	(default) An independent verifier recomputes OCI → logicalRootfsDigest and signs an attestation naming the OCI digest, function id, logicalRootfsDigest, verifier identity, time, and toolchain version.
node-derived	The node re-derives logicalRootfsDigest from OCI layers before first use; strongest, most expensive; SHOULD run off the TTFE critical path. A recomputed digest differing from the declared one MUST reject and emit a security event.

3.2 Canonical logical rootfs tree encoding (LLT1)

logicalRootfsDigest = terrapin-sha256(LLT1-encode(canonical tree)). JSON or any other interchange form MUST NOT affect the digest; the LLT1 byte stream is the only input to the Terrapin computation.

Primitive encodings (shared with LLAY1, §4.3):

- PRIM-1: u8/u16/u32/u64 are unsigned little-endian, fixed width.
- PRIM-2: i64 is signed two's-complement little-endian (used for mtime seconds).
- PRIM-3: varbytes = a length prefix followed by exactly that many raw bytes; path, symlink target, and xattr key/value use a u32 length prefix.
- PRIM-4: all multi-byte integers are little-endian regardless of host.
- PRIM-5: a 32-byte digest field is the RAW Terrapin identifier: the 32-byte G(manifest) payload itself, never the "terrapin-sha256:<hex>" string and never the bare recursive tree root. ("raw Terrapin identifier" / "raw TerrapinID" mean exactly this throughout §3 and §4.)

Document framing:

```
magic      4 bytes = "LLT1" (0x4C 0x4C 0x54 0x31)
version    u16 = 1
entryCount u64
entries     entryCount EntryRecords, sorted
```

- LLT1-D0C-1: entryCount MUST equal the number of EntryRecords, AND the byte stream MUST end exactly after the last entry; a count mismatch or trailing bytes MUST reject (length framing detects truncation).
- LLT1-D0C-2: entries MUST be sorted by normalized path bytes, bitwise lexicographic, directories and files in one global order; the root directory (empty path) sorts first.
- LLT1-D0C-3: normalized paths MUST be unique within the tree; two entries with the same normalized path MUST reject (materialization resolves same-path collisions per MAT-8 before encoding; the encoded LLT1 tree is already collision-free).

Entry record, common prefix (every entry, this exact order/width):

```
pathLen u32; path (pathLen bytes, normalized); type u8 (1=dir 2=regfile
3=symlink 4=chardev 5=blockdev 6=fifo); mode u32 (low 12 bits significant);
uid u32; gid u32; mtimeSec i64; mtimeNsec u32; hardlinkGroup u64 (0 = not
hardlinked); xattrCount u32; xattrs (xattrCount XattrRecords, sorted by key).
```

Type-specific tail (immediately after the prefix):

```
regfile  : contentLen u64; contentId 32 bytes (raw TerrapinID of the content
           object, §4.1: not hex, not the bare tree root)
symlink  : targetLen u32; target (raw link bytes, unresolved)
chardev  : major u32; minor u32
blockdev : major u32; minor u32
dir/fifo : (no tail)
```

```
LLT1-ENTRY-1: fields MUST appear in exactly this order and width.
LLT1-ENTRY-2: type outside 1..6 MUST reject; sockets and any other type are
               unrepresentable and MUST reject.
LLT1-ENTRY-3: mtimeNsec >= 1_000_000_000 MUST reject.
LLT1-ENTRY-4: for regfiles, contentLen MUST equal the content object's length
               and contentId MUST be its 32-byte TerrapinID.
LLT1-ENTRY-5: mode MUST be (st_mode & 0o7777): perms + setuid/setgid/sticky;
               type bits MUST be masked off (type is carried separately).
LLT1-ENTRY-6: a hardlink materializes as a regfile sharing the target's content
               object AND inode metadata (mode/uid/gid/mtime/xattrs); only path
               and hardlinkGroup distinguish members.
```

XattrRecord: keyLen u32; key; valLen u32; val.

```
LLT1-XATTR-1: XattrRecords MUST be sorted by key bytes, bitwise lexicographic.
LLT1-XATTR-2: keys MUST be unique within an entry; a duplicate key (including
               duplicate PAX records resolving to the same key) MUST reject.
LLT1-XATTR-3: an xattr key is derived from a PAX "SCHILY.xattr.<name>" record by
               stripping exactly the literal prefix "SCHILY.xattr."; the stored
               key is the remaining bytes <name> verbatim (the full namespaced
               attribute name, e.g. "user.foo", "security.capability"), with no
               Unicode normalization. An empty key MUST reject. No namespace
               filtering is applied at the identity layer (every namespace is
               represented as raw key bytes); namespace policy, if any, is an
               admission concern, not part of logicalRootfsDigest.
```

Path normalization (exactly MAT-6/MAT-7): raw bytes, no Unicode normalization; relative to root (no leading "/", no trailing "/", single "/" separators); drop "." components and collapse "/" without altering valid component bytes; reject absolute paths, NUL bytes, and ANY "." component. The root is the single empty-path type-1 entry (attributes from a "." tar entry if present, else defaults mode 0o755, uid 0, gid 0, mtime 0, no xattrs).

Hardlink groups: members share contentId and a non-zero hardlinkGroup; ids are assigned deterministically in sorted path order starting at 1 (the lowest-path member of each set gets the next id); non-hardlinked entries use 0; a hardlink to an absent target MUST reject.

Whiteouts/opaque/sparse: the canonical tree is the RESULT of applying layers; markers MUST NOT appear as entries; sparse follows §4 (hole structure is derived canonically from content and does not affect contentId; source-archive sparse metadata is non-portable and MUST NOT be used for identity).

Tar/PAX corner cases: PAX extended headers override ustar fields when present for path, linkpath, uid, gid, size, mtime, and SCHILY.xattr.*; GNU/PAX sparse headers are NOT used for identity; tar type map: '0'/'\0'→regfile, '5'→dir, '2'→symlink, '1'→hardlink, '3'→chardev, '4'→blockdev, '6'→fifo, '7'→regfile, any other → reject. The tar parser MUST preserve raw path/linkname bytes including embedded NUL; truncating at NUL before the NUL rejection (MAT-6) fires is a path-confusion hazard.

PAX numeric and timestamp canonicalization (these feed logicalRootfsDigest, so they MUST be exact):

```
LLT1-NUM-1: integer PAX values (uid, gid, size) are base-10 ASCII. Parsing MUST
             accept an optional leading '-' only where the field is signed,
             accept leading zeros, and otherwise require digits [0-9] only; any
             other byte MUST reject. uid/gid MUST fit u32 and size MUST fit u64,
             else reject (overflow MUST NOT wrap or saturate).
LLT1-NUM-2: uid and gid are unsigned; a negative value MUST reject. size MUST be
             non-negative and MUST equal the content object's logical length.
LLT1-TIME-1: mtime is a base-10 decimal "seconds[fraction]". The integer part
             is parsed into mtimeSec as a two's-complement i64 (negative,
             pre-1970, allowed); overflow MUST reject.
```

- LLT1-TIME-2: the fraction is interpreted at nanosecond resolution by FLOORING:
 mtimeSec = floor(value) and mtimeNsec = round-toward-zero of
 (value - floor(value)) * 1e9, so mtimeNsec is always in
 [0, 1e9). Fractional digits beyond 9 are truncated, not rounded. A
 value with no fraction yields mtimeNsec = 0. (Flooring makes
 negative timestamps unambiguous: "-1.5" → mtimeSec -2,
 mtimeNsec 500000000.)
- LLT1-NUM-3: ustar octal fields are used only when no PAX override is present;
 they follow standard ustar octal parsing and the same range checks.
-

4. Filesystem plane: content objects and the physical layout (LLAY1)

The logical rootfs (§3) is realized physically as content objects under the content-addressed layout. This is the read path's source of filesystem bytes and the unit of cross-image dedup.

4.1 Content objects

- C0-1: Every logical regular file MUST have exactly one content-object identifier (§2.3) over its content bytes, and in v1 those bytes are stored as a CAS content object under that identifier. (Inlining file bytes into the base descriptor is reserved, not used in v1: SMALL-2.)
- C0-2: A content object's identifier MUST be terrapin-sha256 over the file's logical (hole-expanded) content bytes, independent of image, layer, path, mode, owner, mtime, or xattrs.
- C0-3: Two files with identical content MUST resolve to the same content-object identifier; within a cache domain they MUST resolve to the same CAS entry, proof blocks, and warmth credit.
- C0-4: Hardlinked files MUST reference the same content object; hardlink group identity is a property of the logical tree (§3), not the content object.
- C0-5: A content object is blocked from offset 0 at the fixed 2 MiB size. A file smaller than one block is a single short block, so a cold read of a small file fetches only that file's bytes. A zero-length file has no data blocks and resolves to the empty content object, whose Terrapin identifier is the canonical manifest digest for length 0 (with tree root G(empty)); the bare tree root is never the identifier (§2.2).
- C0-6: Content-object construction MUST be deterministic.
- C0-7: Sub-file content-defined chunking MUST NOT be used; it is a reserved future layout and MUST NOT change the content-object identity defined here.

Sparse files and zero blocks:

- SPARSE-1: The content-object identifier MUST be over hole-expanded logical bytes; hole structure MUST NOT affect the content-object identifier.
- SPARSE-2: Hole structure MUST be derived canonically from content (a 2 MiB-block-aligned all-zero region is canonically a hole), not from the source archive's sparse encoding.
- SPARSE-3: The canonical zero block is an all-zero 2 MiB Terrapin block; its leaf G is the constant in §2.2/§15. This constant is a VERIFICATION value, not a stored CAS object: fully-zero block-aligned ranges are never stored (SPARSE-4) and are represented as virtual-zero ranges (SPARSE-7). A node holds at most the single constant, never per-occurrence copies.
- SPARSE-4: Physical storage MUST NOT materialize hole/zero blocks; the layout references the canonical zero block for fully-zero ranges.
- SPARSE-5: A zero-hashing shortcut MUST produce the identical content-object identifier that full hole-expanded hashing would (preserves MAT-20).
- SPARSE-6: Runtime SEEK_HOLE/SEEK_DATA follows the canonical hole structure.
- SPARSE-7: A canonical zero range is a virtual verified-zero range derived from (zeroLeafDigest, length, cache domain); it requires no stored bytes and creates no ordinary object-root CAS identity, lease, or warmth.
- SPARSE-8: Virtual zero applies to PROOF blocks too, not just data. A hash-file/proof block (level >= 1) whose entries are fully determined by canonical zero subtrees MAY be represented virtually: it MUST NOT be stored or fetched, and MUST verify byte-identically to the materialized Terrapin hash-file block it

stands for. This is required for very large sparse objects (notably the guest-address-linear memory image, MEM-1, where a high-address mapped-data block's proof siblings may be entirely zero subtrees): a verifier resolves such proof blocks from the zero constant (§2.2), never the network. A conformance vector **MUST** exercise a high-address mapped-data region behind a large zero span (§15.10 ENC-CONF-2).

Small files:

- SMALL-1: Every regular file **MUST** be a content object regardless of size (a tiny file is one short block, CO-5), maximizing cross-image dedup of shared small files. Startup latency from many small files is addressed in the transport plane by packs (§7; startup cohesion is reserved in v1, §4.4), **NOT** by changing file identity.
- SMALL-2: Inlining file bytes into the base descriptor is **RESERVED** and **MUST NOT** be used in v1. Rationale: it defeats dedup for the highest-fan-out small files (an inlined stream earns no CAS presence or warmth, working against the density thesis), bloats the hot, must-verify-on-every-restore base descriptor, and couples profiling to base identity (the separation §4.4 COH-3 and LAYOUT-5 establish). If ever reintroduced, inline bytes **MUST** be a bootstrap-bundled Terrapin object that preserves content-object identity and CAS promotion, never raw bytes committed into LLBD1.
- SMALL-3: (Reserved with SMALL-2.) Were inline reintroduced, an inlined file would still be identified by its content-object identifier and verified (terrapi-sha256(bytes) == the committed contentDigest) before exposure (DIGEST-BIND-6).
- SMALL-4: Grouping multiple distinct files into one object **MUST NOT** be used as a dedup mechanism (group identity depends on all members). A startup-cohesion object (reserved in v1, §4.4) would likewise be a transport optimization with no dedup or identity role, not such a grouping.
- SMALL-5: (Reserved with SMALL-2.)

4.2 Physical layout function (content-addressed)

Identifier: llifs-layout-content-addressed-v1. Regular-file data is stored as content objects; the tree metadata (paths, modes, owners, mtimes, xattrs, symlink/device/FIFO nodes, hardlink groups) remains base-descriptor metadata (§3), not deduped as file data; startup ordering is a fetch schedule (§7), not a physical repack.

- LAYOUT-1: A layout **MUST** declare the logicalRootfsDigest it implements.
- LAYOUT-2: A no-hole regular file **MUST** be a single identity extent with objectOffset == fileOffset == 0 and object == the file's content object.
- LAYOUT-3: Sparse files **MUST** be represented per SPARSE (virtual-zero extents, canonical zero block).
- LAYOUT-4: layoutDigest **MUST** commit the full file → content-object → extent mapping.
- LAYOUT-5: layoutDigest **MUST NOT** commit fetch ordering, pack membership, cache state, or peer availability; two deployments with different startup schedules over the same files **MUST** produce the same layoutDigest.

4.3 Canonical physical layout encoding (LLAY1)

layoutDigest = terrapi-sha256(LLAY1-encode(layout)), reusing the §3.2 primitives. All 32-byte digest fields are raw Terrapin identifiers (G(manifest)), never hex and never the bare tree root.

Document framing:

```
magic "LLA1" (0x4C 0x4C 0x41 0x31); version u16 = 1;
layoutFn (varbytes, e.g. "llifs-layout-content-addressed-v1");
implementsLogical 32 bytes (raw TerrapinID of the logicalRootfsDigest);
objectCount u64 + ObjectDescriptors (sorted by digest bytes, unique);
fileCount u64 + FileLayouts (sorted by normalized path bytes);
inlineCount u64 + InlineRecords (sorted by path).
```

(Startup-cohesion objects, reserved in v1 (§4.4), are in any case NOT part of LLAY1 / layoutDigest: were they defined they would be a profiler-driven transport optimization committed by the startup profile/pack (§7), never changing layoutDigest, per LAYOUT-5.)

```
ObjectDescriptor: digest 32; blockSize u64 (2097152); length u64; tree 32;
                  class u8 (1=content object; other values reserved, MUST reject
                  in this layout).
```

FileLayout: pathLen u32; path; size u64; backing u8; then EXACTLY the tail for that backing value and nothing else (no sentinel/placeholder for the non-applicable index):

backing==1 (object): objectIndex u64; extentCount u32; extents.

```
backing==2 (inline, RESERVED in v1, LLAY1-11): inlineIndex u64;
    (extentCount u32 = 0; no extents).
```

backing==3 (empty): (extentCount u32 = 0; no extents; no index).

```
Extent: kind u8 (1=data 2=virtual-zero); objectOffset u64; fileOffset u64;
length u64.
```

```
InlineRecord: pathLen u32; path; contentDigest 32; size u64.
```

LLAY1-1: all counts MUST equal the records that follow, AND the byte stream MUST end exactly after the last record; trailing bytes MUST reject.

LLAY1-2: implements logical MUST equal the LLT1 logicalRootfsDigest; a verifier MUST reject on mismatch.

LLAY1-3: ObjectDescriptor.digest MUST equal G(manifest(blockSize,length,tree));
 blockSize MUST be 2097152; the bare tree root MUST NOT appear as an
 identifier. ObjectDescriptors MUST be sorted by digest bytes and
 unique by digest.

LLAY1-4: for backing==1, extents MUST tile [0,size) exactly (sorted by fileOffset, contiguous, non-overlapping, no gaps, no coverage past size); for content-addressed layout objectOffset MUST equal fileOffset. For backing==2 and backing==3, extentCount MUST be 0.

LLAY1-5: a virtual-zero extent (kind==2) MUST align to the object's zero-block leaves, consume no stored bytes, and set objectOffset == fileOffset (the same identity-mapping rule as data extents; there is no separate object backing for zero ranges).

LLAY1-6: backing==3 (empty file) has size 0 and no extents; hardlinked files share one objectIndex.

LLAY1-7: layoutDigest MUST commit layoutFn, implementsLogical, all object descriptors, all file layouts/extents, and all inline records, and nothing about fetch order, pack/cohesion membership, cache state, or peer availability (LAYOUT-5).

LLAY1-8: unknown values of backing, Extent.kind, or ObjectDescriptor.class MUST reject. FileLayout paths and InlineRecord paths MUST each be unique and sorted by normalized path bytes; an inline path MUST NOT duplicate an object-backed file path.

LLAY1-9: every index MUST reference an existing record of the exact required class: `FileLayout.objectIndex` → an `ObjectDescriptor` with `class==1`; `FileLayout.inlineIndex` → an `InlineRecord`. An out-of-range or wrong-class index MUST reject.

LLAY1-10: every ObjectDescriptor MUST be referenced by at least one FileLayout.objectIndex, and every InlineRecord by at least one FileLayout.inlineIndex; unreferenced descriptors or inline records MUST reject (no unreachable records may perturb layoutDigest).

LLAY1-11: Inline backing is RESERVED in v1 (SMALL-2): inlineCount MUST be 0, no InlineRecords may be present, and backing==2 MUST reject. The inlineCount field, the InlineRecord layout, and backing==2 are retained for forward-compatibility only.

4.4 Startup cohesion (reserved in v1)

A startup-cohesion object would concatenate startup-critical small-file bytes into fewer, larger blocks to cut per-block startup overhead. It is RESERVED and MUST NOT be emitted in v1: serving file bytes from a concatenated object requires a committed segment map (content-object id, file offset, cohesion object id, cohesion offset, length) and per-segment rules proving each served segment against its original content-object identity, which v1 does not define. The v1 mechanism for startup small-file fragmentation is the PACK (\$7): a pack batches the startup working set of canonical content-object blocks into one fetch, with no alternate byte carrier and

no new identity.

- COH-1: Startup cohesion MUST NOT be emitted in v1. If introduced later it MUST be a named structural object (LLCOH1) with a canonical segment map and explicit rules that every served segment verifies against its original content-object identity before exposure (DIGEST-BIND-6).
- COH-2: A cohesion object, if ever defined, MUST NOT receive dedup or cross-image warmth credit and MUST be additive only: the same content MUST remain addressable by its content-object identifier, never sole-sourced from it.
- COH-3: A cohesion object, if ever defined, MUST NOT be part of physical layout identity (MUST NOT appear in LLAY1 / layoutDigest, §4.3) and MUST be committed by the startup profile/pack (§7), so it never changes layoutDigest (LAYOUT-5).

4.5 Mount model and copy-up

imagefsd serves a read-only lower filesystem; the writable upper is OverlayFS, outside the read-only base. The default mount is FUSE; for gVisor the lower is served through a CAS-backed lisafs gofer with no host-kernel FUSE on the path (§11).

- MOUNT-1: Mount MUST NOT require full hydration unless materialization strictness requires it.
- MOUNT-2: Directory traversal and stat for startup-critical paths MUST be served from local bootstrap metadata before STARTUP_READY.
- MOUNT-3: The writable upper is outside the read-only base. While the agent runs the upper is an ordinary OverlayFS upper and is not verified by LLIFS. On suspend, the upper's diff MUST be captured into the filesystem delta (§6) as a canonical overlay-delta object; once captured it is read-only and Terrapin-verified like any other object.

Copy-up is a latency hazard: a one-byte write to a large lower file can force a large synchronous read. Trusted runtime requirements (not untrusted profiles) MAY mark expected-mutable files with a strategy:

prehydrate-full-file	fetch+verify the whole file before STARTUP_READY
seed-upper	synthesize the file into the upper before start
volume-required	redirect the path to writable storage (admission may reject otherwise)
deny	fail fast on mutation
allow-lazy-below-threshold	permit lazy copy-up only below a size

- COPYUP-1: Lowerdir mutation MUST be treated as a latency hazard.
- COPYUP-2: Mandatory copy-up behavior MUST come from trusted runtime requirements, not from an untrusted profile.
- COPYUP-3: A prehydrate-full-file path MUST be fetched and verified before STARTUP_READY; a volume-required large path SHOULD cause admission to reject unless redirected to writable storage.
- COPYUP-4: Copy-up full-file fetches MUST be measured separately from ordinary foreground reads.

5. Memory plane: base snapshot, copy-on-write restore, restore hazards

The filesystem plane (§§3-4) delivers verified, deduped, lazily-faulted files. The memory plane delivers the same for guest memory: a verified, deduped, lazily-faulted base memory snapshot from which sandboxes restore, sharing the base copy-on-write so a million agents are one shared base plus a small per-agent overlay. This section defines the shared base and the restore protocol; the persisted memory delta and the lifecycle that mints it are §6.

5.1 Base memory snapshot

A base memory snapshot is captured once from a warmed sandbox (interpreter, framework, and client loaded; paused at a restore point) and committed by a base descriptor (§2.3, §2.5). The base descriptor commits: the base memory snapshot OBJECT (the guest memory image, a Terrapin object, §5.3); the runtime-state blob (non-memory sandbox state: threads, fds, namespaces, vCPU/sentry state, needed to resume); and the memory layout (guest-address → object-offset mapping, §5.3). A restore working set (§5.5) is associated PROFILE metadata keyed to the base version, not a committed base-descriptor field.

- MEM-1: The base memory snapshot object identifier MUST be terrapin-sha256 over the canonical HOLE-EXPANDED, guest-address-linear byte image of the committed guest address space [0, committedEnd), committedEnd being the end of the highest region in the memory layout (§15.5 ENC-ML-1). Byte offset 0 in this image is guest offset 0 (MLAYOUT-1). Mapped-data ranges contribute their captured bytes; mapped-zero and unmapped/no-access ranges contribute canonical zeros. Identity is over LOGICAL bytes, computed with the canonical zero-block shortcut (the §4 SPARSE rule on the memory plane), so it is independent of physical sparseness and never materializes zero ranges.
- MEM-2: The base is captured once per base version, hashed, and signed. Producers are NOT required to reproducibly regenerate an identical image; identity is "snapshot once, content-address the result." (Contrast MAT-20: the filesystem base is reproducibly derivable; the memory base is captured.)
- MEM-3: A base memory snapshot is executable state and MUST NOT be restored unless reachable from a trusted, signed base descriptor (DIGEST-BIND-3/6).
- MEM-4: In addition to the full base-descriptor binding of DIGEST-BIND-3, the runtime-state blob and memory layout MUST be committed by the base descriptor so the (memory + state + layout) triple cannot be recombined across bases.
- MEM-5: The MEM-1 identity commits LOGICAL bytes and does NOT encode mapping state: a mapped-zero range and an unmapped/no-access range hash identically (both zeros). Per-range mapping STATE (§5.3 MLAYOUT-5) is committed ONLY by the signed memory layout, which the base descriptor binds (MEM-4, DIGEST-BIND-3). Restore MUST derive every range's state from the signed layout and MUST NOT infer state from snapshot bytes; two snapshots with identical base_memory but different layout STATE are different bases.

5.2 Storage granularity (the block/page decoupling)

This restates §2.4 normatively for the memory plane.

- GRAN-1: The base memory snapshot object MUST use the standard 2 MiB profile; the 2 MiB block is the unit of fetch and verification into the node CAS only.
- GRAN-2: The unit of per-agent copy-on-write sharing MUST be the platform MMU page (4 KiB on x86-64; the native page size otherwise). CoW granularity is independent of block size.
- GRAN-3: Once a 2 MiB base block is verified-resident in the node CAS, its pages MUST be shareable copy-on-write across sandboxes without re-fetch and without re-verification.
- GRAN-4: Implementations MUST NOT introduce a sub-2-MiB memory profile unless measured cold-node over-fetch (§5.5) justifies it; if justified, a finer profile MAY be defined separately and MUST be reported in telemetry. Page dedup MUST NOT be the justification; dedup is achieved by base sharing (§5.4), not by content-addressing each agent's pages.

5.3 Memory layout (guest-address-linear)

Memory addresses are load-bearing (pointers); pages MUST NOT be permuted in the stored object the way file bytes may be repacked (§4). The base memory snapshot object is stored in guest-address-linear order, which is both canonical for dedup and trivially MAP_PRIVATE-mappable as one linear region.

- MLAYOUT-1: Guest-address-linear: in the canonical hole-expanded image (MEM-1) byte offset 0 is guest offset 0 across the whole committed space

[0, committedEnd), not merely within one region. Physical storage MAY be sparse (holes for zero, unmapped, and not-yet-fetched ranges); logical offsets are unaffected.

- MLAYOUT-2: A base memory snapshot is exactly ONE Terrapin object (§2.3); multiple objects MUST NOT be used. Its IDENTITY is the hole-expanded logical byte image of MEM-1; its PHYSICAL storage is sparse. The memory layout describes the guest regions and per-range state over that one object. Mapped-data ranges occupy stored object bytes (resident only after fetch and verify). Mapped-zero and unmapped/no-access ranges are logical holes: they contribute canonical zeros to identity (MEM-1) but occupy NO stored or transferred bytes, and they keep their guest offsets (NOT compacted out) so the image remains one linear MAP_PRIVATE region (MLAYOUT-1). The layout MUST be deterministic.
- MLAYOUT-3: Restore ordering MUST be expressed only as a fetch schedule (§5.5), never as a permutation of the stored object (MLAYOUT-1).
- MLAYOUT-4: The memory layout MUST commit the guest-region → object-extent mapping and the page size assumed at capture.
- MLAYOUT-5: The memory layout MUST commit, per guest address range, its exact SIGNED mapping STATE (one of three), and restore MUST reproduce it. Cache residency (resident vs not-yet-fetched) is an ORTHOGONAL runtime condition, NOT a signed state:
- mapped-data: backed by base object bytes. When not yet resident, a touch traps, fetches, and verifies before exposure (COW-6); when resident, pages are shared copy-on-write (COW-1). "Absent" is this state's not-yet-resident runtime condition, not a fourth state.
 - mapped-zero: the canonical verified zero block (§4 SPARSE-3); a touch installs a verified zero page (COW-6b), no fetch.
 - unmapped / no-access: restore as UNMAPPED/protected. It MUST NOT become mapped-zero and MUST NOT be lazily fetched; a touch MUST fault to the guest exactly as it would natively, never silently resolving to a page.
- The layout MUST also commit page protections and mapping attributes (R/W/X, guard pages, shared vs private), and restore MUST reestablish them.

5.4 Copy-on-write restore protocol

The node CAS materializes the base memory snapshot object as a sparse memfd/file: verified 2 MiB blocks are written at their layout offsets; not-yet-fetched blocks are holes. Each restored sandbox maps the base region MAP_PRIVATE over that file; the kernel shares resident base pages across all sandboxes and copy-on-writes a private page on first write. Per-agent dirty pages are the overlay (§6).

Sharing one verified base across sandboxes via MAP_PRIVATE requires the runtime's memory backing to support a shared, verified base file; this is the one net-new runtime capability and is specified technically in §11.

CRITICAL: for memory, zero is valid data. A hole meaning "block absent (not yet fetched)" MUST NEVER be confused with a page that legitimately reads zero. An unprotected hole resolves to a zero-filled page on touch, silently exposing unverified, incorrect memory. Every absent base range MUST be trapped (COW-6) before any sandbox can touch it.

- COW-1: Restore MUST back the guest base region with a MAP_PRIVATE mapping of the verified base file, so resident base pages are physically shared and writes fault to private copies.
- COW-2: Per-agent dirty pages are the memory overlay (§6); they MUST NOT be written to the shared CAS and receive no dedup or warmth credit (they are ephemeral until persisted as a memory delta, §6).
- COW-3: A node MUST hold at most one physical copy of a given base block's resident pages per cache domain, shared across all sandboxes restoring that base.
- COW-4: Restore MUST NOT require loading the full base image; only the restore working set (eager) plus lazily-faulted pages (COW-6) are required to resume (the memory analog of MOUNT-1).
- COW-5: Implementations SHOULD cap the per-agent dirty-page set and report agents whose overlay growth defeats density (so they can be migrated or rescheduled).

Verification:

- MVERIFY-1: Every base memory byte MUST be Terrapin-verified at CAS admission, before it is ever mapped into any sandbox.
- MVERIFY-2: Verification is per-block-per-node and amortized: once a base block is verified-resident, per-agent CoW sharing of its pages MUST NOT re-verify (GRAN-3). Verification cost does not scale with agent count or fault count.
- MVERIFY-3: Per-agent dirty pages are agent-generated and MUST NOT be verified against any Terrapin identifier; they are not CAS content.

Fault handling:

- COW-6: Absent base ranges MUST be protected so they cannot resolve to zero-filled or stale pages. First touch MUST synchronously fault through the LLIFS handler, fetch and verify the containing block at P0, ADMIT the verified block into the node CAS / shared base backing, then map or continue the faulting page from that verified shared backing (preserving the shared-base invariant COW-1/COW-3, never a per-sandbox unverified copy), and only then resume the faulting thread. A read of an absent range MUST block, never return zeros.
- COW-6a: Absent (not-yet-fetched) ranges require userfaultfd MISSING-fault handling (populate verified content before resuming), NOT minor-fault handling. MISSING handling MAY use UFFDIO_COPY only where it populates, or continues from, the shared verified base backing (COW-1/COW-3); it MUST NOT install an unverified or per-sandbox-private copy that bypasses the shared base. Minor-fault / UFFDIO_CONTINUE applies only to ranges already present and verified in the shared base backing that are being mapped into an additional sandbox. Implementations MUST choose the fault mode by backing state, not assume one.
- COW-6b: A base block the signed memory layout designates as the canonical zero block (§4 SPARSE-3) is verified content, NOT absent: the handler MAY install a verified zero page immediately (e.g. UFFDIO_ZEROPAGE) without a fetch. "Known-zero per signed layout" MUST be distinguished from "absent" (COW-6) and MUST be driven by the signed layout, never inferred from a memfd hole.
- COW-7: A hard failure to fetch or verify a faulted base block MUST surface as a distinct, process-fatal-class event (cf. §8), not a silent hang, and MUST follow the §8 fetch-fallback and hedging rules.

5.5 Restore working set (prehydration)

Cold-node restore cost is dominated by fetching the base blocks the guest touches during resume. A restore working set lists those blocks, prioritized, to prehydrate before resume; the cold tail faults lazily (§5.4). This is the memory-plane analog of the startup pack; the full profile/pack/coverage machinery is §7.

- MRESTORE-1: A restore working set MUST reference base memory snapshot blocks as canonical BlockRefs (object / level / blockIndex).
- MRESTORE-2: imagefsd SHOULD prehydrate the working-set blocks into the node CAS before resume, within configured byte and latency budgets.
- MRESTORE-3: Blocks not in the working set MUST be faultable lazily during execution (COW-6) at foreground (P0) priority.
- MRESTORE-4: A restore working set MUST be associated with a base version and a memory profile, not only a base object digest.
- MRESTORE-5: imagefsd MUST report cold-node memory over-fetch (bytes fetched but not touched in the measured restore window); this is the signal that decides whether a finer memory profile (GRAN-4) is warranted.

5.6 Restore hazards

Restoring N agents from one base clones state that MUST NOT be shared. These hooks are mandatory, not optional.

- RHAZARD-1: Entropy MUST be refreshed on restore (kernel RNG, userspace PRNG

- seeds, cached nonces/keys derived at capture). Reusing base entropy across agents is a security defect.
- RHAZARD-2: Time sources MUST be refreshed (monotonic and wall clocks jump relative to the base).
- RHAZARD-3: External state captured in the base (open sockets, host fds, in-flight connections, leases) MUST be treated as dead and reconnected or invalidated.
- RHAZARD-4: The base capture point SHOULD precede any agent-, tenant-, or request-specific secret entering memory, so the shared base holds no cross-agent-sensitive data. For multi-tenant or cross-request reuse this strengthens to MUST, or capture-time secret scanning/attestation is required.
- RHAZARD-5: Restore-hazard handling MUST be part of signed base/runtime policy; a base lacking required hooks MUST NOT be used for multi-tenant restore.
- RHAZARD-6: Cross-tenant base sharing MUST obey cache domains (§9); a memory base is a content-presence and behavioral side channel and MUST NOT be shared across domains unless explicitly opted in.
- RHAZARD-7: Before HAZARDS_CLEARED for multi-tenant restore, each of the following MUST be refreshed/invalidated or asserted not-applicable by signed policy: PID/TID and cached process identity; futexes, robust lists, rseq, TLS; vDSO/vvar time mappings; timers, epoll/poll, pending signals, pending async I/O; ASLR / address-space assumptions (shared base ⇒ shared layout; inject new per-agent randomness post-restore, RHAZARD-1); page protections and mapping attributes (R/W/X, guard pages, no-access ranges, shared vs private) reestablished per MLAYOUT-5; seccomp, namespaces, cgroups, credentials, capabilities; language-runtime state (Go scheduler/netpoller, JVM safepoints, Python hash seed, OpenSSL/BoringSSL DRBG reseed); capture-time secret scanning or a no-secrets-before-capture guarantee (RHAZARD-4).

5.7 Memory-restore lifecycle

```

BASE_UNRESOLVED
→ BASE_RESOLVED          (signed base descriptor verified)
→ BASE_STARTUP_PREHYDRATED (restore working set resident and verified)
→ RESTORED                (sandbox mapped CoW; base runtime-state then
                           runtime delta applied, RDELTA-5)
→ HAZARDS_CLEARED        (RHAZARD-1..7 refreshed)
→ RUNNING

```

Failure states: BASE_DESCRIPTOR_UNTRUSTED, BASE_MEMORY_BLOCK_UNAVAILABLE, BASE_VERIFICATION_FAILED, RESTORE_HAZARD_HOOK_MISSING, COW_BACKING_UNAVAILABLE.

6. Agent state and lifecycle

An agent is a shared, read-only BASE (base rootfs §4 + base memory snapshot §5, bound by a base descriptor §2.5) plus a per-agent OVERLAY across memory and filesystem. This section defines how the overlay is persisted (the delta), how a snapshot is identified (the State Root) and branched (lineage and fork), how a snapshot is written, and the RAM-reclamation mechanisms (reset, hibernate, yield) with their storage tiers. It builds directly on §4 (content objects, the writable upper) and §5 (base memory snapshot, copy-on-write restore, restore hazards).

6.1 Overlay and delta

While an agent runs, its per-agent state is an OVERLAY (§2.3): copy-on-write dirty guest-memory pages (§5.4) plus the writable filesystem upper (§4.5). The overlay is ephemeral; it lives in RAM and node-local scratch, is never written to the shared CAS, and earns no dedup or warmth (COW-2, MOUNT-3).

At suspend, hibernate, or fork, the overlay is captured into a DELTA (§2.3): a

persisted, content-addressed, per-agent object with three parts (memory, filesystem, and runtime).

- DELTA-1: The MEMORY DELTA MUST be a page MAP: for each diverged guest page, the source of its bytes (this delta's own data object, an ancestor's, the base, or the canonical zero) and range. The map directly resolves every page, so resume needs no ancestor traversal (§6.3 LINEAGE-3) and a fork shares parent/base pages by direct reference with no copy. Pages identical to the base remain shared from the base copy-on-write (§5.4) and need not be listed. Its byte encoding is §15 (LLMD1).
- DELTA-2: The FILESYSTEM DELTA MUST be the canonical overlay-delta object (§2.3, §15): content-object references for changed regular-file data, plus every overlay namespace and metadata operation: created/changed directories, symlinks, device and FIFO nodes, ownership/mode/mtime/xattr changes, renames, and deletions (whiteouts/tombstones). It MUST be sufficient to reconstruct the writable upper exactly. Hardlink identity in the upper is NOT preserved across copy-up (as in default OverlayFS): copied-up files are recorded by content, not as a hardlink group; restoring hardlink identity is a future extension.
- DELTA-2R: The RUNTIME DELTA captures the agent's diverged NON-memory, non-filesystem runtime state: the runtime-native execution state the base runtime-state blob (§5.1) no longer describes after the agent has run, and which guest memory alone does not reconstruct (for gVisor e.g. thread/register state, fd tables and offsets, epoll/timer/signal/futex/rseq/TLS state, syscall-restart state, VMA metadata, socket state, and sentry/kernel bookkeeping). A memory delta plus filesystem delta cannot restore a non-cooperating process hibernated mid-request or yielded while blocked upstream; the runtime delta is what makes that exact. For v1 it is an OPAQUE runtime-native Terrapin object, like the base runtime-state blob; LLIFS defines no internal structure (§15 LLRD1). Its encoding MAY later be optimized into a delta against the base runtime-state blob; correctness comes first.
- DELTA-3: A delta is per-agent and private: it MUST live in per-cache-domain-keyed CAS and MUST NOT be deduped across cache domains (§9). Each delta component is a Terrapin object verified before exposure (DIGEST-BIND-6).
- DELTA-4: Capturing the overlay MUST occur under the single freeze barrier (WRITE-1) so the memory, filesystem, and runtime deltas share one freeze epoch; restore MUST present a coherent memory+filesystem+runtime view (no mmap'd working-plane page may reflect a different epoch than the memory delta).
- DELTA-5: A persisted delta (memory, filesystem, or runtime) MUST be directly resolvable on the resume critical path: every object it needs is named by absolute Terrapin id and read directly, with NO ancestor or lineage traversal (LINEAGE-3, §6.3 FORK-3). Storage MAY share ancestor objects zero-copy, and a named ancestor OBJECT (the base via base_ref, or an LLMD1 source==2 ancestor delta-data object) is a DIRECT resolution input read without traversal. The lineage LINKS proper, the parent State Root and an LLFD1 parent_ref, are provenance, authorization, GC, or compaction inputs ONLY; the resume-critical resolved form (direct component object ids) MUST be obtainable without walking lineage (e.g. via the bootstrap, §7 BOOT-2).
- RDELTA-1: A State Root that persists the memory plane MUST also commit a runtime delta, UNLESS signed runtime policy proves no non-memory runtime state is required for resume. reset, golden, clean, and filesystem-only persistence need none (§6.6).
- RDELTA-2: The runtime delta is a Terrapin object over runtime-native bytes (§15 LLRD1), interpretable ONLY by a runtime satisfying the sandbox pin (§2.5) and the compatibility matrix (§11.4); LLIFS defines no portable structure for it in v1.
- RDELTA-3: The runtime delta MUST be captured under the SAME freeze barrier as the memory and filesystem deltas (DELTA-4, WRITE-1), sharing one freeze epoch.
- RDELTA-4: The runtime delta MUST be directly resolvable on the resume critical path with no lineage traversal (DELTA-5).
- RDELTA-5: Restore MUST apply the base runtime-state blob THEN the runtime delta before unfreeze, and MUST fail closed on a mismatch or on a missing required runtime delta (RDELTA-1).

6.2 State Root

A State Root (§2.3) is the content-addressed identity of one persisted agent snapshot. It is minted at suspend, hibernate, or fork, never for a merely-running

agent. Its canonical byte encoding and identifier (a Terrapin object) are §15; this section fixes the committed fields and their rules.

A State Root commits:

actor	stable agent identity; survives across all snapshots of one agent. A fork allocates a NEW actor (§6.3).
parent	the State Root this derives from, or "none" for genesis; the lineage edge (§6.3).
base	the base descriptor reference (§2.5): the shared base rootfs + base memory snapshot + runtime-state blob + memory layout this snapshot restores onto.
memory	the memory delta reference (DELTA-1), or "none".
filesystem	the filesystem delta reference (DELTA-2), or "none".
runtime	the runtime delta reference (DELTA-2R, RDELTA-1), or "none"; required when the memory plane is persisted unless signed policy proves none is needed (RDELTA-1).
sandbox pin	the per-architecture sandbox pin (§2.5), required when memory is persisted.
workload	the workload identity (SR-5), required when any per-agent plane (memory or filesystem) is persisted.
run_mode	which planes persist (§6.6).
producer	the producing node/control-plane identity; the runtime-attestation subject (§2.5).
created	production time (descriptive).

- SR-1: A State Root MUST commit all fields above; absent components MUST be the explicit "none", never omitted. Its identifier MUST be the Terrapin object identifier of its canonical encoding (§15); there is no separate digest scheme (§2.1).
- SR-2: A State Root MUST reference a base descriptor; restore MUST validate the State Root under the runtime regime AND the base descriptor under the builder regime (DIGEST-BIND-4/6); a trusted State Root does not confer trust on its base.
- SR-3: memory and filesystem are the per-agent state references (the runtime delta accompanies a persisted memory plane per SR-4/RDELTA-1); setting NEITHER memory nor filesystem is valid (a clean or golden-only snapshot, §6.6). A resume reads whichever references are non-"none".
- SR-4: A State Root that persists ANY per-agent plane (memory or filesystem) MUST commit the workload identity (SR-5); a delta is workload-specific and MUST NOT be restored against a different workload. A State Root that persists the memory plane MUST additionally commit the sandbox pin (§2.5, DIGEST-BIND-7) and the runtime delta unless signed policy proves none is required (RDELTA-1).
- SR-5: The WORKLOAD IDENTITY is a canonical Terrapin digest (§15 LLWI1) over the RESTORE-VISIBLE fields of the control plane's workload spec (atelet.WorkloadSpec): every field that, if different at restore, could invalidate or make unsafe the reapplication of a persisted memory/filesystem delta. It MUST commit, at pod scope: the pause image, hostname, the host- and shared-namespaces flags, and the pod security context; and per container, in declared order: name, image, command, args, env, working directory, run-as user/group, capabilities (added and dropped), security profile (privileged, read-only-rootfs, no-new-privileges, seccomp/AppArmor/SELinux), mount and volume-device topology (mount path, sub-path, read-only, source name/type), exposed devices, and the restore-relevant resource limits (at least the memory limit, which constrains the guest address space). Fields intentionally EXCLUDED (they do not affect restored state or are regenerated post-restore) are enumerated in §15 LLWI1 (ENC-WI-2). On restore the supplied workload MUST match the committed workload identity exactly; a mismatch MUST reject, never silently restore. Any atelet.WorkloadSpec field that affects restore correctness MUST be added to LLWI1 and to this list, never silently ignored.
- SR-6: created is committed, so it participates in identity: two snapshots of one actor collapse to one State Root identifier ONLY when all committed fields, including created, are byte-identical (intended idempotent-retry behavior, not a collision). Snapshots that differ only in created are distinct State Roots; an implementation wanting retries to collapse across a time boundary MUST reuse the original created value.
-

6.3 Lineage and forking

The parent State Root reference forms a hash-linked DAG over State Roots: a State Root identifier transitively commits to its ancestry. A FORK creates a new actor that begins from an existing snapshot's exact state, then diverges: the differentiator for speculative agent branching.

- FORK-1: A fork MUST allocate a NEW actor identity and a new State Root whose parent is the source snapshot.
- FORK-2: base (the base descriptor) MUST be shared by reference, never copied.
- FORK-3: The child's memory, filesystem, and runtime deltas MUST begin as copy-on-write references to the source State Root's delta components; only diverged pages/files MAY be newly stored. No per-plane copy is required at fork time. These references MUST be DIRECT and self-contained: a child delta names the exact base/ancestor component objects it depends on, so resume resolves them by direct reference and never walks the lineage chain (LINEAGE-3). A compaction step MAY flatten a chain into direct references, but resume MUST NOT depend on traversal.
- FORK-4: A fork MUST NOT require rehydrating the source onto the child's node before the child can resume; cold pages/blocks fault on demand (§5.4, §8). A fork is a read fan-out of one snapshot to N children and SHOULD reuse the fan-out, hedge, and seed machinery of §8/§10.
- FORK-6: Self-containment (DELTA-5) moves cost off the resume path onto snapshot and compaction: producing a self-contained child MAY require O(parent-dirty-pages + parent-upper-metadata) work (a large LLMD1 map naming ancestor pages directly; a complete LLFD1 upper). This is an intentional tradeoff. For high-fan-out speculative branching, a later Merkle-map/trie representation of LLMD1/LLFD1 MAY let forked children structurally share maps while staying directly resolvable; v1 accepts the flat cost.
- FORK-5: A fork's State Root MUST be runtime-attested by the node/control plane performing it (§2.5); the verifiable parent link preserves the chain to the source. Fork authorization MUST be a control-plane policy decision (who may fork actor X); the authorization hook MUST exist. (A deployment MAY relax enforcement; phase guidance is §16.)
- LINEAGE-1: parent MUST reference a resolvable, verifiable State Root or be "none".
- LINEAGE-2: The substrate MUST be able to enumerate a State Root's lineage for GC (§9) and revocation (§14).
- LINEAGE-3: Lineage depth MUST NOT be on the resume critical path: a resume reads the State Root's direct base/memory/filesystem/runtime references, not the ancestor chain (§7).

6.4 Snapshot write/commit path

Agents PRODUCE state; the write path is normative. On suspend, hibernate, or fork:

1. Freeze the agent (quiesce the runtime).
 2. Checkpoint memory; compute the page-indexed memory delta against the base (and parent, for a fork), writing only changed pages (DELTA-1). Capture the runtime delta (the diverged non-memory runtime state, DELTA-2R) unless signed policy proves none is required (RDELTA-1).
 3. Capture the writable upper into the filesystem delta (DELTA-2).
 4. Write all new objects to the CAS under temporary names and atomically promote each only after Terrapin verification.
 5. Compute the memory, filesystem, and runtime delta object identifiers; assemble and identify the State Root (§15).
 6. Obtain the runtime attestation over the State Root (§2.5).
 7. Publish the State Root to the control plane; demote node-pinned objects toward the cold tier per the run mode and tier policy (§6.6, §9).
- WRITE-1: Steps 1-3 MUST occur under a single freeze barrier so the memory, filesystem, and runtime deltas share one freeze epoch (DELTA-4, RDELTA-3, §5).
 - WRITE-2: New objects MUST be written under temporary names and atomically promoted only after Terrapin verification.
 - WRITE-3: A State Root MUST NOT be published until every object it references is durably present in at least one tier the resume path can reach.
 - WRITE-4: The freeze-publish path MUST be crash-safe: a crash before publish MUST

leave the prior State Root valid and resumable; partial objects MUST be GC-able garbage (§9), never referenced.

WRITE-5: Snapshot-write latency MUST be reported separately from resume latency (§13).

6.5 Reclamation mechanisms: reset, hibernate, yield

Farm-to-LLM agents are memory-bound and mostly idle (blocked on LLM calls, between turns, or done). The substrate reclaims their RAM via three mechanisms, all of which are §5 restores differing only in which delta is reapplied:

```
reset      stateless completion → restore the base with NO delta.
hibernate  stateful pause → restore base + the agent's persisted delta.
yield      reclaim during a blocking upstream (LLM) call → hibernate a blocked
           agent + an upstream transaction that survives the eviction.
```

DESIGN TENET (non-cooperative baseline): agents run arbitrary libraries that will not cooperate. Cooperation OPTIMIZES; OBSERVATION guarantees. Correctness MUST hold for fully non-cooperating workloads.

ROBUST-1: Cooperation (completion signals, yield hints, an SDK) is an OPTIMIZATION; correctness and acceptable reclamation MUST hold for non-cooperating workloads, which MUST NEVER be corrupted.

ROBUST-2: The substrate observes guest state because the sandbox kernel owns the guest memory, filesystem, and network stack, and an L7 ingress/egress proxy owns the request connection and the upstream call. Together they let the substrate reclaim non-cooperating agents.

ROBUST-3: A completed successful request/response pair marks a candidate request BOUNDARY and MAY trigger reclamation; it does NOT prove the process is safe to reuse (a 2xx can leave leaked secrets, mutated globals, or background work). Therefore the DEFAULT reclamation on a boundary is restore-fresh (RESET-1); failed/errored pairs MUST restore-fresh.

RESET-1: The default reset is restore-fresh: bind the next request to a freshly restored base instance and discard the completed one. The fresh instance shares resident base pages copy-on-write (§5.4), so it is cheap (no per-agent delta to load), NOT a cold start.

RESET-2: Reset MUST reset EVERY mutable plane: guest memory, the filesystem upper, fd table/offsets, task/thread/timer/signal/futex state, sentry/kernel runtime state, and external resources (RHAZARD), not merely drop the memory overlay.

RESET-3: Reset MUST clear RHAZARD-1..7 (§5.6) before the instance serves, or assert each not-applicable by signed base policy.

RESET-4: In-place rewind (rewinding a running process's full state to base in its existing slot) is a DEFERRED optimization, permitted only under a machine-checkable quiescence predicate AND a measured win over restore-fresh; until then the substrate uses restore-fresh.

HIBER-1: A hibernation snapshot captures only the per-agent delta over the immutable base (the base is not copied); wake = §5 restore + delta apply.

HIBER-2: The delta MUST be content-addressed and Terrapin-verified; wake verifies it before exposure (DELTA-3).

HIBER-3: Wake MUST clear RHAZARD-1..7 (reseed entropy, refresh time, reconnect or invalidate external state).

HIBER-4: Eviction policy (idle timeout, memory pressure, explicit yield) MUST be trusted policy, not agent-controlled.

HIBER-5: Wake is on the event's critical path at P0 with a wake-latency budget; wake latency MUST be reported (§13).

HIBER-6: Hibernation deltas are leased objects (§9); the base refcount MUST reflect every agent (active or hibernated) that will wake to it.

YIELD-1: Yield is hibernate applied to an agent blocked on an upstream call, plus an upstream transaction. The proxy MUST take durable ownership of the upstream request before the agent is evictable; the pre-ack window MUST NOT be evictable and pre-ack eviction attempts MUST be observable.

YIELD-2: Upstream SUBMISSION MUST be single-flight under a durable transaction id bound to (agent invocation, upstream target, request-body digest, retry epoch), issued and durably logged BEFORE submission begins. The proxy MUST NOT, on its own initiative, submit the same transaction id more than once,

and the id MUST NOT deduplicate distinct invocations.

- YIELD-2a: Single-flight submission does NOT by itself guarantee at-most-once upstream EXECUTION. The proxy's durable log MUST model each transaction as one of not_sent, sent_unknown (submitted, outcome unconfirmed: crash or ambiguous network failure), response_held (a definitive response captured), or definitive_failure. From sent_unknown the proxy MUST NOT silently resubmit a non-idempotent call. At-most-once EXECUTION across a sent_unknown ambiguity is achievable ONLY with upstream cooperation: an idempotency key the upstream deduplicates on, or a cooperative upstream transaction protocol. Absent that, per-upstream policy MUST choose between failing definitively (no resubmission) and resubmitting (accepting possible duplicate execution); the substrate MUST NOT claim general exactly-once effects for arbitrary non-idempotent upstreams.
- YIELD-3: The agent-proxy downstream connection is NOT preserved (RHAZARD-3); on wake the restored delta resumes its pending read and the proxy delivers the single held response (replay for retrying clients; a defined continuation or a defined client-visible error otherwise), under a bounded hold timeout.

State machine:

RUNNING

→ (pair done; default & on failure) RESET (restore-fresh) → base → RUNNING
 → (stateful pause) HIBERNATE → delta → ... → WAKE
 → (blocking upstream call) YIELD = hibernate(delta) + txn → WAKE
 All paths pass through RHAZARD-1.7 clearance before serving.

6.6 Storage tiers and run modes

Two orthogonal axes govern a persisted snapshot: the RECLAMATION MECHANISM (§6.5) and the storage TIER of its delta.

Tier (where a delta lives) maps to the control plane's existing actor states:

local (PAUSED) the delta's objects are pinned on the node VM(s) for sub-second wake if rescheduled there; the worker allocation is released (compute is reclaimed) but the bytes stay local. The warm-pool state.

external (SUSPENDED) the delta lives only in object storage; no node RAM/SSD reservation and no worker. The "costs nothing while idle" state. Wake re-acquires through the tiered cache (§9), preferring a peer/local copy before object storage.

- TIER-1: The tier MUST be resolved from which CAS/tier holds the State Root's objects; it is reported via control-plane actor state, not a competing per-snapshot field (§12).
- TIER-2: local→external promotion/demotion MUST be an explicit, retryable, crash-safe transition: while a delta is promoting to external it remains validly local (resumable), local objects MUST NOT be evicted until the external copy is durable (WRITE-3), and a crash mid-promotion MUST leave the agent resumable, not stateless.

Run modes (the run_mode field, SR-1) select which planes persist across a suspend/stop:

clean boot from the base with no per-agent state (memory and filesystem "none").

golden start from a shared base memory snapshot (plus base rootfs) with per-agent state discarded on stop; sharing falls out of base copy-on-write (§5). ("golden" here names this RUN MODE only; it is not an object class: the "golden memory" object term is retired, §17.)

persist-rootfs the filesystem delta persists; memory is "none" (warm fs, cold process).

persist-rootfs+memory the filesystem, memory, and runtime deltas persist (exact process resume requires the runtime delta, RDELTA-1); the flagship sub-second wake.

RUNMODE-1: run_mode is the authoritative selector of which planes a State Root persists; a request to discard a plane MUST NOT silently preserve it.

RUNMODE-2: run modes are orthogonal to tier (any run mode may be local or external) and to the reclamation mechanism.

6.7 Density model (informative)

Reclamation drives the per-agent overlay toward zero, but node-resident RAM also includes substrate overhead that does not. The agent-overlay term:

```
agent_resident = base + sum over ACTIVE agents of overlay
  reset:    completed → 0 agent overlay (no stored delta; discarded)
  hibernate: paused   → ~0 agent RAM (delta on local/external tier)
  yield:    blocked   → ~0 agent RAM (delta on tier; the proxy holds the
    in-flight call: relocated, not eliminated)
```

Total node-resident RAM also includes the per-sandbox runtime floor (sentry/runtime + gofer + netstack buffers), filesystem overlay residency, proxy-held request/response buffers, and snapshot/restore working memory. Resident concurrency is therefore runtime-floor-bound, while TOTAL hosted agents is bounded by ACTIVE concurrency plus delta tier capacity, so for mostly-blocked LLM agents, reset/hibernate/yield are load-bearing for density, not merely nice to have: they free the runtime floor, which base sharing alone cannot.

7. Startup and resume

Both cold start (an OCI image's first exec) and warm resume (a State Root's restore) deliver only the bytes needed to begin useful execution, then fault the rest. The machinery is shared: a weakly-trusted PROFILE optimizes (what to fetch first), trusted RUNTIME REQUIREMENTS compel (what must be present), a PACK is the transport envelope, a BOOTSTRAP carries the metadata+proofs so the critical path needs no metadata round-trip, and COVERAGE measures readiness. Startup (cold) and resume (warm) are duals throughout.

7.1 Profiles (performance hints)

A startup profile (cold) or resume profile (warm) is a weakly-trusted hint recording the working set faulted earliest. Profiles optimize; they MUST NOT compel (that is §7.2).

- PROFILE-1: A profile MUST NOT weaken verification or alter any identity. It orders and predicts; it never authorizes.
- PROFILE-2: A startup profile MUST be selected using the final command, args, working directory, and explicitly whitelisted env keys, never arbitrary user-controlled env, which MUST NOT select an expensive mandatory action.
- PROFILE-3: A profile MUST be keyed by the identities it optimizes, plus the selector and a profile id, not by a mutable tag. The cold key is (logicalRootfsDigest, layoutDigest, selector, profile id). The warm key is (base-descriptor digest, State-Root digest, profile id); the restore working set within it is additionally keyed by base version and memory profile (§5.5).
- PROFILE-4: Runtime-observed foreground faults MUST override profile order.
- PROFILE-5: A resume profile records the pages/files a wake touches first; for the memory plane it is the restore working set (§5.5).

7.2 Runtime requirements (trusted policy)

Runtime requirements are trusted policy derived from signed metadata, admission

policy, or operator configuration, never from an untrusted profile alone.

- POLICY-1: Mandatory prehydration, fail-closed behavior, pod rejection, expected-mutable-file handling (§4.5), and mmap/exec-critical coverage MUST come from runtime requirements, not a profile.
- POLICY-2: imagefsd MAY use profile hints for bounded prefetch but MUST NOT exceed configured byte/CPU/network/latency budgets unless trusted policy authorizes it.
- POLICY-3: Requirements affecting fail-closed behavior MUST be covered by signature or attestation policy (§14).

7.3 Packs (transport envelopes)

A pack, startup pack (cold) or resume pack (warm), is a transport envelope, not a mounted object. It carries canonical block payloads that are scatter-extracted into the canonical CAS, so pack bytes never form a separate cache identity.

- PACK-1: Pack payloads MUST be addressed by canonical BlockRefs (object / level / blockIndex).
- PACK-2: After fetching a pack, imagefsd MUST scatter-extract verified blocks into the canonical CAS; runtime reads MUST be served from canonical CAS keys, never a separate pack namespace.
- PACK-3: Duplicate delivery of a block through multiple packs MUST NOT create duplicate cache identity or duplicate warmth credit.
- PACK-4: A pack MUST be bound to the corresponding profile key (PROFILE-3): the cold key for a startup pack, the warm key for a resume pack.
- PACK-5: A pack MUST NOT be required for correctness unless trusted policy requires it; absent a required pack, behavior is governed by §7.2 and the failure rules of §8.
- PACK-6: Packs MAY be zstd-compressed; compressed bytes MUST be verified (wire digest) or bounded-and-sandboxed before decompression. A pack is transport, not an object: it carries no Terrapin identity of its own. Each block it delivers MUST verify against its canonical BlockRef / Terrapin proof before CAS admission and exposure (§8); the decompressed payload as a whole is not a Terrapin object.

7.4 Bootstrap (metadata + proofs, no critical-path round trip)

A bootstrap is the single compact metadata record (not a Terrapin object) that makes a start begin with zero metadata round-trips: a startup bootstrap for cold start, a resume bootstrap for warm resume. A bootstrap MUST carry not only fetch metadata but ALL trust and validation inputs the critical path needs, so verification itself requires no round trip (BOOT-3).

- BOOT-1: A startup bootstrap MUST carry the metadata needed to reach a mountable, startup-ready filesystem AND to validate it without a round trip: root inode and startup-critical path trie, endpoint/loader metadata, the selected profile's BlockRef list, the Terrapin manifests for startup-required objects, the inlined proof material to verify them, and the trust material for the cold path's signature/policy and materialization-assurance checks (the signed base descriptor and its assurance record, §3, §14), or proof these are already locally cached.
- BOOT-2: A resume bootstrap MUST carry everything needed to validate AND restore with no round trip: the State Root object; the runtime attestation envelope (§2.5); the referenced base descriptor and its builder-regime validation material (signature/attestation, sandbox pin, memory layout) for SR-2 / DIGEST-BIND-6; the memory, filesystem, and runtime delta references resolved to their DIRECT component objects (§6.3); the restore working set (§5.5) and resume pack BlockRef list; and inlined proofs for all resume-critical blocks. Any of these MAY be omitted only if proven already locally cached and verified.
- BOOT-3: P0 startup/resume reads AND their verification MUST NOT require a metadata, proof, or attestation network round trip; everything needed before exec/unfreeze MUST be in the bootstrap or locally cached. (Proof blocks for an object below a configurable size SHOULD be inlined or prefetched whole;

one 2 MiB level-1 block authenticates up to 128 GiB, §2.2.)

BOOT-4: A bootstrap SHOULD be prewarmed/seeded to candidate nodes: startup bootstraps to seed nodes for a rollout, resume bootstraps to candidate resume nodes for PAUSED actors and seed nodes for SUSPENDED actors (§10).

7.5 Coverage

Coverage measures how much of a required set is present on a node, over canonical BlockRefs (and, for cross-image base sharing, over content objects, §10).

COVERAGE-1: Coverage and scheduler warmth MUST be computed over canonical BlockRefs (block presence), not over packs.

COVERAGE-2: A pack receives warmth credit only for the canonical blocks it delivers into CAS; duplicate bytes MUST NOT inflate coverage.

COVERAGE-3: Proof-block coverage MUST be tracked separately from data-block coverage.

COVERAGE-4: Coverage MUST be cache-domain-aware (§9).

7.6 Time-to-first-exec and time-to-resume

TTFE (cold) and TTR (warm) are the primary latency SLOs and MUST be broken into components so operators can attribute a slow start.

TTFE = image resolve + metadata/bootstrap resolve + signature/policy verify + materialization-assurance verify + profile select + startup pack fetch + scatter/verify + mount + runtime create + exec

TTR = State Root resolve + runtime-attestation verify + base resolve (warm via the base plane) + memory, filesystem & runtime delta resolve + verify + resume pack fetch (hot working set) + restore (page-in; base runtime-state + runtime delta applied) + overlay attach + unfreeze + first request

TTFE-1: Metrics MUST decompose TTFE and TTR into the components above, so a start can be attributed (metadata-, proof-, network-, decompression-, verification-, restore-, policy-, or runtime-bound) (§13).

TTR-1: A resume MUST fault the hot working set first and lazily fault the cold tail at P0 (§5.4, §8); full hydration MUST NOT be on the critical path unless run mode or policy requires it. If the runtime restore eagerly touches the whole memory image, the start MUST be reported as restore-bound, not as a thin lazy resume. The runtime delta is resolved directly (RDELTA-4) and applied with the base runtime-state before unfreeze (RDELTA-5); it is part of restore, not lazily faulted.

7.7 Profile drift and materialization strictness

DRIFT-1: imagefsd MUST report profile misses (P0 startup/resume faults not covered by the selected pack), by phase.

DRIFT-2: A profile SHOULD be marked stale when its miss ratio exceeds a configurable threshold (default 20% over a meaningful window) and SHOULD trigger a re-profiling event; a new profile MUST use a new profile identifier and MUST NOT mutate the old profile record. (Profiles and runtime requirements are signed metadata records: their trust identity is the signature/digest over their bytes, §14, not Terrapin object classes.)

Materialization strictness selects how much must be present before a milestone; it is trusted policy and is orthogonal to run mode (§6.6) and tier:

startup-only	only exec/startup-critical present before exec
readiness-critical	exec + readiness present before readiness
full-before-ready	full hydration before readiness
full-before-exec	full hydration before exec
lazy-risk-accepted	runtime faults on cold blocks allowed by policy

STRICT-1: A deployment MUST be able to choose a materialization strictness mode.
 STRICT-2: Strictness mode MUST be visible in status, events, and metrics; runtime lazy-materialization failures MUST be reported with clarity equivalent to a classic image-pull failure (§13).

8. Fetch, verification, and transport

Every byte exposed to a workload is verified against a trusted Terrapin identifier (§2). This section is the read path beneath that guarantee: how a block is located, deduplicated in flight, prioritized, hedged, and sourced (from local cache, peers, mirrors, or origin) without ever trusting transport.

8.1 Read path

cache hit: read → map (path/offset or guest address) → BlockRef → local CAS hit (already verified; local integrity per READ-7) → return bytes.
 cache miss: read → BlockRef → join/create singleflight → P0 fetch → source select → fetch (compressed or raw) → verify the wire digest if present → bounded decompression if needed → load proofs (from bootstrap/cache) → verify the extracted block against its canonical BlockRef / Terrapin proof → admit to canonical CAS → return bytes.
 (Uncompressed-identity verification can only occur AFTER decompression; compressed bytes are verified beforehand only against a wire digest, COMP-2/COMP-3.)

READ-1: Foreground reads MUST have strict priority over background hydration.
 READ-2: Bytes MUST NOT be returned until verified against a trusted Terrapin identifier (§2.2 TERRAPIN-4); this holds for filesystem blocks, memory blocks, and proof blocks alike.
 READ-3: imagefsd MUST coalesce concurrent requests for the same BlockRef unless intentionally hedging (§8.4).
 READ-4: Adjacent readahead SHOULD occur only when it does not violate foreground latency budgets.
 READ-5: Per-container, per-tenant, and per-node in-flight fetch memory MUST be bounded.
 READ-6: A failed source fetch MUST fall back to other allowed sources; a verification failure MUST reject the source and retry elsewhere if policy allows (§8.6, §14).
 READ-7: A local CAS entry is treated as verified after admission (READ-2), so the node's local store (RAM CAS, page cache, on-disk CAS) is inside the node's trusted computing base for already-admitted blocks. A deployment MUST protect that store against post-admission corruption (disk/bit rot, stale page cache, buggy local mutation) by at least one mechanism: a per-block integrity check on read, fs-verity/dm-verity over the CAS backing, or periodic scrubbing against the Terrapin tree. On detected corruption the entry MUST be treated as ABSENT and re-fetched and re-verified (§8.2), never served. (Cross-host transport is never a trust anchor, TERRAPIN-4; READ-7 is about the LOCAL store after admission.) The mechanism is an operational choice, but one MUST be stated and enabled.

8.2 Block singleflight

Terrapin verifies 2 MiB blocks; a single sequential read may shatter into many smaller VFS/fault requests. A per-BlockRef singleflight state machine collapses them.

block lifecycle: ABSENT → INFLIGHT → VERIFIED → (EVICTED) ; → FAILED

BLOCK-1: imagefsd MUST maintain a single in-flight operation per BlockRef.
 BLOCK-2: Multiple reads/faults for ranges within the same block MUST join the same in-flight operation.

BLOCK-3: The first miss MAY trigger a full 2 MiB fetch; later reads MUST NOT trigger duplicate range requests unless intentionally hedging (§8.4).
 BLOCK-4: Once verified, all waiters MAY be satisfied from RAM CAS or page cache.
 BLOCK-5: Singleflight state MUST be cache-domain-aware (§9).

8.3 Fetch priority

P0 synchronous foreground read or page fault (workload is blocked)
 P1 predicted next foreground read (strong evidence: sequential, mmap/ELF locality, recent trace, profile)
 P2 selected profile / pack critical block
 P3 readiness-path prefetch
 P4 background hydration
 P5 full-image archival / prewarm

QUEUE-1: P0 MUST preempt P2-P5.
 QUEUE-2: P1 SHOULD be generated only from strong evidence.
 QUEUE-3: P4/P5 MUST be rate-limited during cluster-wide rollout.
 QUEUE-4: Fetch admission SHOULD consider NIC pressure, disk-queue depth, registry rate limits, peer health, cache-domain policy, and origin breaker budgets.
 QUEUE-5: Background hydration MUST be interruptible.

8.4 Hedged requests and origin circuit breakers

Hedging cuts individual fault tail latency; origin breakers stop hedges from stampeding a degraded origin during a correlated rollout. Both are required.

HEDGE-1: P0 requests MUST support hedged fetch; a hedge SHOULD issue when the first source exceeds $\min(\text{configured hedge delay}, \text{remaining fault budget} / 4)$.
 HEDGE-2: The first VERIFIED response wins; losing hedges SHOULD be cancelled immediately.
 HEDGE-3: A corrupt response MUST penalize or eject the source; a slow response MUST lower its score but MUST NOT be treated as corrupt.
 HEDGE-4: P2-P5 background hydration SHOULD NOT hedge by default.
 HEDGE-5: Hedging MUST respect tenant, registry-auth, egress, and cache-domain policy.
 ORIGIN-1: P0 origin hedging MUST be protected by per-registry and per-image circuit breakers and a token-bucket (or equivalent) origin-hedge budget.
 ORIGIN-2: When the origin breaker is open, P0 hedges MAY target local mirrors or peers but MUST NOT stampede origin unless emergency policy allows it.
 ORIGIN-3: The controller SHOULD coordinate origin-hedge budgets cluster-wide during rollouts; origin-hedge denials MUST be visible in telemetry (§13).

8.5 Local cache hierarchy and source selection

Recommended fetch order: process page cache → imagefsd RAM CAS → local SSD/NVMe CAS → same-host cache → same-rack/AZ peer → node-local registry mirror → regional mirror/CDN → origin registry.

SRC-1: Foreground reads MUST bypass or preempt background hydration at every tier.
 SRC-2: For P0, source selection SHOULD optimize expected completion time (queue delay + tail latency + verification-failure + cross-AZ + auth-refresh + origin-breaker penalties), not raw bandwidth; background hydration MAY prefer high throughput.
 SRC-3: Startup/mmap-critical and proof blocks SHOULD be retained preferentially in cache (§9).
 SRC-4: Cache entries MUST be promoted atomically only after verification, and keyed in the internal digest plane with cache-domain separation (§9).

8.6 Peer protocol

Peers accelerate fan-out but are never trusted for integrity. Peers exchange Terrapin BlockRefs and proof blocks, not OCI blobs.

- PEER-1: Peers MUST NOT be trusted for integrity; receivers MUST verify every block locally before exposure (READ-2).
- PEER-2: Peers MAY refuse service on load/tenant/network pressure and MUST apply rate limits to avoid cascading failure.
- PEER-3: Cross-cache-domain Have()/FetchBlock() MUST be denied unless policy explicitly allows it (§9); registry credentials MUST never be sent to peers.
- PEER-4: Corrupt peer responses MUST eject/penalize the peer and MUST be observable (§13).
- PEER-5: P2P is not required for correctness; the protocol is defined so it can be added without changing internal identity (phasing is §16).

8.7 Registry and mirror interaction

LLIFS uses OCI registries for external storage and discovery: registry-facing operations use OCI sha256 descriptors; internal verification uses terrapin-sha256.

- REG-1: OCI descriptor verification MUST use the external digest plane; filesystem/memory-byte verification MUST use the internal plane.
- REG-2: LLIFS MUST support origin-registry fallback unless policy forbids it, and SHOULD prefer mirrors / pull-through caches near the cluster.
- REG-3: LLIFS SHOULD use HTTP range requests where supported and MUST preserve registry authentication boundaries.
- REG-4: LLIFS SHOULD discover metadata via OCI Referrers, and MUST support a tag-fallback discovery mode for registries without Referrers; tag-fallback records MUST still bind to the OCI image digest internally, and signature/trust verification MUST be identical under both.

8.8 Compression and decompressor hardening

LLIFS distinguishes filesystem/memory identity (uncompressed) from wire identity (compressed).

- COMP-1: An LLIFS object's identity MUST be the Terrapin digest of its uncompressed canonical bytes (§2.1 DIGEST-2, §4, §5). A wire (compressed) digest is transport-only and MAY additionally be carried; it MUST NOT be used as object identity.
- COMP-2: If a wire digest is available, compressed bytes MUST be verified before decompression; otherwise decompression MUST enforce bounded compressed input, output, memory, CPU/time, and expected length, in an isolated decompressor.
- COMP-3: Decompressed output MUST be verified against the uncompressed identity (per block / per object) before exposure; unverified decompressed bytes MUST NEVER reach VFS or guest memory.
- COMP-4: Decompression, archive parsing, and untrusted-metadata parsing SHOULD run in isolated workers with seccomp, rlimits, no ambient network, no registry credentials, and a minimal filesystem view.

8.9 Failure behavior

- FAIL-1: If a foreground (P0) block cannot be fetched-and-verified from any allowed source, the read/fault MUST fail (and, for a memory fault, surface as the process-fatal-class event of COW-7), never hang silently or expose zeros.
- FAIL-2: Repeated verification failure for one object SHOULD trigger security alerting (§14).
- FAIL-3: If imagefsd crashes, active mounts SHOULD recover from verified cache state

or fail closed.

FAIL-4: Failures that would classically appear as ImagePull errors MUST be reported with equivalent clarity when they occur during runtime lazy materialization (§13, STRICT-2).

9. Cache domains, leases, and garbage collection

The fetch tiers and source order are §8.5. This section governs WHO may share a cached block (cache domains and keying), how shared objects are kept alive (leases and liveness), and how they are reclaimed (GC), at agent-FaaS scale, where one base object is referenced by a million agents.

9.1 Cache domains

A cache domain is the maximum scope within which content-presence timing leakage is acceptable (§2, §17). It is the isolation unit for dedup, keying, warmth, and GC.

DOMAIN-1: A cache domain has a canonical ID: a length-prefixed raw byte string derived from a configured source (cluster | node-pool | namespace | tenant | workload-identity | enclave-group). The ID is opaque bytes; no Unicode normalization.

DOMAIN-2: The domain source MUST be configured per deployment and MUST be deterministic: the same workload context maps to the same domain ID.

DOMAIN-3: Domain assignment MUST be trusted (admission/snapshotter from the workload's namespace/tenant/identity), never workload-controlled. A workload MUST NOT choose its own domain to reach another's cache.

Keying (two modes; §8.5 SRC-4):

```
trusted-shared:  cache_key = terrapin-sha256:<object> / level / blockIndex
private-domain:  cache_key = HMAC(domainKey[epoch], <object> / level / blockIndex)
```

KEY-1: In private-domain mode, physical cache names MUST be the HMAC of the canonical BlockRef under a per-domain secret, so content presence is not inferable across domains by probing cache names.

KEY-2: The per-domain secret MUST be node-local and MUST NOT be sent to peers or registries (PEER-3).

KEY-3: Keys are VERSIONED by epoch. Rotation bumps the epoch and writes new entries under the new key; rotation MUST NOT delete or rewrite existing entries (avoiding a mass invalidation storm); old-epoch entries become unreferenced and are reclaimed by GC.

KEY-4: A bounded grace window MAY keep the prior epoch readable. A prior-epoch entry is GC-eligible ONLY when it has no live lease (LEASE-3); an old-epoch entry that still has a live lease MUST remain readable, or be lazily re-written under the current epoch, before the grace window expires, so rotation never strands live-leased content (rotation MUST NOT reduce reachability of any leased object).

KEY-5: Proof-block caches obey the same domain keying as data blocks. Shared proof caching MAY be enabled only under the same cache-domain policy gate as cross-domain content sharing (SHARE-1, REDACT-3), because proof-block presence also reveals object presence.

9.2 Leases and liveness

Dedup means one content object, base object, or delta may be referenced by many images, profiles, or agents within a domain.

Lease types: runtime, startup, profile, peer, operator pin, cache-domain, materialization-attestation, pause (pins a PAUSED actor's objects on the local tier), and state-root (pins a SUSPENDED State Root's objects durably without a

node pin).

- LEASE-1: The substrate MUST maintain liveness per (cache domain, object).
Acquiring any referencing lease pins the object; releasing unpins it.
- LEASE-2: Lease updates MUST be atomic and crash-safe: a crash MUST NOT leak a reference (pinning forever) nor drop one (evicting in-use content). Liveness MUST be reconstructable after a crash from a durable journal or by rescanning the live lease set, so neither leaked nor dropped pins survive recovery.
- LEASE-3: An object with a live lease in a domain MUST NOT be evicted from that domain; liveness aggregates across ALL referencing images/agents; eviction is a domain-level decision, not per-image.
- LEASE-4: Hibernation deltas are leased objects: a hibernated agent holds a lease on its delta, and the base it will wake to MUST be kept live by a lease that reflects every active AND hibernated agent (§6 HIBER-6).

9.3 Garbage collection

- GC-1: GC MUST be digest-safe and crash-safe: partial objects written under temporary names, atomically promoted only after verification, never deleting an object with a live lease (§6 WRITE-2/WRITE-4).
- GC-2: GC MUST respect cache-domain boundaries; an object's eligibility is evaluated per domain by that domain's liveness and epoch.
- GC-3: Eviction order SHOULD prefer unreferenced, low-fan-in, cold objects, and preferentially retain high-fan-in objects (shared by many images/agents) and startup/mmap-critical/proof/base blocks; evicting a high-fan-in object harms many workloads at once.
- GC-4: Stale-epoch entries (post-rotation, past the grace window) with no live lease are first-class GC candidates.
- GC-5: GC SHOULD export eviction reasons (§13).

Fleet-scale GC (a shared base object may have ~a million referrers, so a hot per-agent recount on it is impractical). GC is ONE model: a mark-sweep whose ROOTS are the live leases (§9.2) and the live State Roots; an object survives iff it is reachable from a root. Leases and State Roots are roots/pins for that sweep; this reconciles LEASE-3 (a leased object is never evicted) with GC-FLEET (no hot per-operation recount): "not per-operation recount" means the base is not incremented/decremented per agent, NOT that leases are ignored.

- GC-FLEET-1: Shared base objects (base roots content objects + the base memory snapshot, including the snapshot used by the golden run mode) MUST be pinned by a template/operator lease while the template is live and MUST NOT be GC'd by per-agent churn; this avoids a hot per-agent recount on the base.
- GC-FLEET-2: Per-agent objects (memory, filesystem, and runtime delta components) are retained iff REACHABLE from a live actor's retained State Root or a live lease, plus lineage retention/squash (§6.3), not by a per-object recount.
- GC-FLEET-3: The mark-sweep from each live State Root MUST traverse its base descriptor reference, so a base stays live while ANY retained State Root (active or hibernated) can resume from it, even if the template/operator lease has been removed (resolving LEASE-4 vs GC-FLEET-1). The sweep MUST NOT use per-operation recounting and MUST respect cache-domain boundaries (GC-2).
- GC-FLEET-4: Lineage retention policy MAY squash intermediate State Roots in a suspend chain (merging deltas into a later State Root); squashing MUST preserve the resumability of retained State Roots and the self-contained direct references of §6.3.

9.4 Multi-tenancy and redaction

- REDACT-1: Metrics exposed to a tenant MUST be scoped to that tenant's domain and MUST NOT reveal other domains' image presence, cache warmth, peer availability, or content-presence timing.
- REDACT-2: Cross-domain aggregates (total cache size, global warmth, dedup ratios) MAY be exposed only to the operator/cluster role.

REDACT-3: Base-warmth and content-presence signals used for scheduling (§10) MUST be domain-scoped in any tenant-visible surface; cross-domain warmth is operator-only.

9.5 Cross-domain sharing (opt-in)

SHARE-1: Cross-domain content sharing (trusted-shared mode, unkeyed) MUST be explicit opt-in; the default is private-domain (keyed). Cross-domain Have()/FetchBlock() are denied unless policy allows (PEER-3).

SHARE-2: A deployment MAY define a trust hierarchy (e.g. share within a tenant's node-pool but not across tenants) by choice of the domain ID source (DOMAIN-1); sharing is within a domain by construction.

10. Scheduler integration and warmth

Keeping the cluster scheduler out of the hot path is a core thesis: placement is guided by WARMTH (what verified state a node already holds) so a start or resume faults few bytes. Warmth is computed over canonical BlockRefs and content objects (§7.5), is cache-domain-scoped (§9), and is exposed to scheduling as a score.

10.1 Warmth model

WARM-1: Warmth MUST be computed over canonical BlockRefs (block presence) and, for cross-image/cross-agent sharing, over content objects and base objects, never over packs (COVERAGE-1/2).

WARM-2: A node MUST be able to advertise BASE warmth (the set of held content objects and base memory snapshot blocks) scoped to a cache domain and INDEPENDENT of any specific image or agent. At agent-FaaS scale base warmth is the dominant placement signal: a node warm on a template's base can run almost any of that template's agents.

WARM-3: Proof-block warmth MUST be tracked separately from data-block warmth.

WARM-4: Per-image / per-profile coverage (§7.5) MUST still be tracked; base warmth is additive context, not a replacement.

WARM-5: All warmth advertisements MUST be cache-domain scoped and MUST NOT reveal presence across domains (REDACT-3); cross-domain warmth is operator-only.

10.2 Node score

A node's score for placing a start/resume SHOULD combine the warmth that reduces its critical path, minus pressure that raises it:

```
score = w1.startup/resume-critical blocks present
      + w2.proof blocks present
      + w3.mmap/exec-critical present
      + w4.base warmth (content objects + base memory snapshot)
      + w5.readiness blocks present
      + w6.peer locality
      - w7.disk pressure - w8.network pressure - w9.origin-breaker risk
```

SCORE-1: All WARMTH inputs to the score MUST be cache-domain-scoped (WARM-5). The node-pressure and origin-breaker-risk terms are permitted non-warmth inputs and MUST themselves be domain-safe (they MUST NOT reveal another domain's content presence).

SCORE-2: Weights are deployment-tunable; the scheduler SHOULD prefer placing latency-sensitive workloads on nodes with high selected-profile and base warmth.

10.3 Resume affinity

- AFFINITY-1: For a PAUSED actor (local tier, §6.6) the scheduler SHOULD prefer the node(s) that still hold the actor's local delta and base resident, for sub-second resume; this is the warm-pool placement signal.
- AFFINITY-2: For a SUSPENDED actor (external tier) the scheduler SHOULD prefer any node warm on the actor's base (WARM-2), since the per-agent delta is small and faults lazily (§7.6).
- AFFINITY-3: Resume affinity MUST be advisory: a State Root MUST be resumable on any node that can validate it and satisfy the sandbox pin (§2.5), so loss of a warm node never strands an actor; it only costs latency.

10.4 Rollout and fan-out

- ROLLOUT-1: For a large rollout, the controller SHOULD resolve image/metadata once per cluster/region, select seed nodes per rack/AZ/cache domain, and prewarm the bootstrap and pack to seed nodes before scheduling pods onto startup-warm nodes (§7.4 B00T-4).
- ROLLOUT-2: A FORK (§6.3) is an intra-cluster rollout of one snapshot to N children; it SHOULD reuse the seed/prewarm/hedge machinery (FORK-4), seeding the source's resume bootstrap and hot blocks to the children's nodes.
- ROLLOUT-3: Background prewarm/hydration MUST be interruptible and rate-limited (QUEUE-3/QUEUE-5) and MUST respect origin-breaker budgets (§8.4).

10.5 Warmth state

- WARM-6: Warmth state exposed for scheduling (per node: startup/proof/mmap/readiness/base coverage ratios, last-verified time) MUST be keyed by the identities it describes: the PROFILE-3 cold key (logicalRootfsDigest, layoutDigest, selector, profile id) for images; the base descriptor digest for bases, within a cache domain, and is carried by the control-plane integration (§12), not by a workload-visible surface (REDACT-1).

11. Runtime adapters

LLIFS is one verified, content-addressed store; runtimes are ADAPTERS over it. Internal identity (Terrapin) and the verify-before-expose invariant do not change per adapter. gVisor (runsc) is the primary target; Firecracker microVM is next.

11.1 Adapter model

- ADAPT-1: A runtime adapter MUST source filesystem and memory bytes from the verified CAS (§8) and MUST NOT expose any byte before it is verified against a trusted Terrapin identifier (§2.2 TERRAPIN-4, §5 MVERIFY-1).
- ADAPT-2: An adapter MUST NOT alter object identity, the binding chain (§2.5), or the restore-hazard obligations (§5.6); these are runtime-independent.
- ADAPT-3: An adapter MUST honor the sandbox pin (§2.5): a snapshot is restorable only under a runtime compatible with the pinned, per-architecture sandbox assets, and the base compatibility matrix (§11.4).

11.2 gVisor (primary)

Most of the memory plane is reused, not built; one capability is net-new.

- GVISOR-1: The filesystem plane SHOULD be served to the sandbox kernel (Sentry) through a CAS-backed lisafs gofer, with no host-kernel FUSE on the read path. Bytes are verified at the serving layer before they reach the guest. This needs no Sentry change.
- GVISOR-2: Lazy/on-demand restore SHOULD be REUSED, not rebuilt: the existing background-restore mode (the runtime starts the workload after kernel state loads and faults remaining pages on demand) is the cold-tail path. The page SOURCE is redirected by serving the runtime's pages file from the same CAS-backed gofer, where bytes are verified as served. The gofer serves the runtime's expected page view by mapping the runtime's pages-file offsets to LLIFS BlockRefs through the committed memory layout (§5.3): the base memory snapshot identity is unchanged; the pages file is a view, not a new object. (If the runtime's native pages-file layout is not already guest-address-linear, the memory layout provides the deterministic mapping.) Lazy, remote-sourced, and verified restore therefore need no Sentry change.
- GVISOR-3: The ONE net-new capability is SHARED COPY-ON-WRITE BASE restore: backing each restored sandbox's base guest-memory region with a MAP_PRIVATE mapping of one shared, verified base memory snapshot so resident base pages are physically shared across sandboxes and only dirty pages are private (§5.4). The runtime's memory-file backing MUST be extended to support mapping each sandbox's base region onto a shared, externally populated, verified base file (today that backing is not pluggable). Without this, every restore materializes its own multi-GiB guest memory and the density thesis fails; with it, a million agents share one base. This is the substrate's sole hard runtime dependency and its central density mechanism. (Justification: §6.7. It is a minimal, surgical change: it changes where base pages are backed, not the checkpoint format.)
- GVISOR-4: The runtime-state blob (§5.1) MUST be compatible with the runtime's platform mode; the mode MUST be recorded in the base descriptor and the compatibility matrix (§11.4).
- GVISOR-5: gVisor directfs (sandbox-direct host-fd filesystem access that bypasses the gofer round trip) MUST be DISABLED for the LLIFS lower in v1: the gofer is the verify-before-expose and cache-domain enforcement choke point (GVISOR-1, §9), and directfs would let the sandbox read lower bytes the gofer never verified. directfs MAY be enabled for the LLIFS lower only once the native path preserves the identical verify-before-expose (DIGEST-BIND-6) and cache-domain (§9) semantics (a §11.5 / Phase 4 evaluation), and MUST NOT be enabled merely for performance.
- GVISOR-6: The byte paths are distinct abstractions even where they share code: rootfs path/offset reads use the CAS-backed lisafs gofer (GVISOR-1); restore page reads use a CAS-backed PAGE PROVIDER (the pages-file view of GVISOR-2); shared base residency is the MemoryFile/shared-backing change (GVISOR-3). A userfaultfd MISSING fault MUST populate the shared verified base backing, not a per-sandbox private copy (COW-6a); once a base range is resident, additional sandboxes MUST resolve it by shared/minor-fault continuation, never by repeating a private copy.

11.3 Firecracker microVM (secondary)

- FC-1: The filesystem plane SHOULD be served via virtiofs backed by the verified CAS.
- FC-2: The memory plane SHOULD use the microVM snapshot/restore path with a userfaultfd handler that serves verified base pages from the CAS, sharing resident pages across microVMs and copy-on-writing per-VM dirty pages (the same §5.4 model).
- FC-3: Firecracker and gVisor adapters MUST address the same base memory snapshot by the same Terrapin identifier; a base captured for one runtime MAY be incompatible with another (§11.4), but the object identity and verification path are identical.

11.4 Base compatibility matrix

A base memory snapshot is restorable only by a compatible runtime. The base descriptor **MUST** commit a compatibility matrix; a node **MUST NOT** restore a base it cannot satisfy.

- MATRIX-1: The matrix **MUST** record: runtime and runtime version; platform mode; CPU architecture and page size; CPU feature mask (a base captured using features absent on the target **MUST NOT** restore there); checkpoint/state format version; sandbox-kernel/guest ABI expectations; and the seccomp/cgroup/namespace assumptions at capture.
- MATRIX-2: A mismatch on any required field **MUST** fail closed with a distinct event (§13), never a silent or best-effort restore. The sandbox pin (§2.5) and the matrix together gate placement (§10 AFFINITY-3).
- MATRIX-3: Compatibility is decided by a named verifier step, MATCH-COMPAT(committed, restorer), where the COMMITTED value records what the producer captured and the RESTORER value is the candidate node's capability: runtime_id and arch **MUST** be equal; platform_mode and page_size **MUST** be equal; runtime_version and checkpoint/state-format version **MUST** satisfy the runtime's declared backward-compat rule (default: equal); the CPU feature mask **MUST** be a superset (the restorer provides every committed feature); and the seccomp/cgroup/namespace and ABI assumptions **MUST** be satisfiable. MATCH-COMPAT **MUST** run before any base or runtime-delta byte is applied (RDELTA-2/5); any unsatisfied field fails closed (MATRIX-2).

11.5 Native verification path (later)

A native, page-cache-integrated backend **MAY** later reduce per-byte overhead; it **MUST NOT** weaken the verification or identity model.

- NATIVE-1: A native backend (e.g. EROFS/fscache, composefs-style verified trees, fs-verity, virtiofs) **MUST** preserve the logical rootfs and layout binding and the base memory snapshot identity.
- NATIVE-2: A native backend **MUST** either verify with Terrapin directly or bind its native verification root (e.g. an fs-verity root) into signed LLIFS metadata, so Terrapin secures network/remote identity while the native mechanism secures local page-cache/disk reads, without double-paying verification on the critical path.
- NATIVE-3: Any native backend **MUST** preserve cache-domain policy (§9).

12. Control-plane binding

LLIFS is a substrate; the agent control plane is its consumer adapter, exactly as containerd/Kubernetes is a consumer adapter for the filesystem plane. This section binds LLIFS to the control plane's existing data model (the ateapi/atelet/ateom protos) rather than introducing a parallel one. Proto additions below are PROPOSED (not the current schema) and are required deliverables, landing in Phase 2 with hibernate/resume (§16).

12.1 Mapping to the existing model

State Root (§6.2)	≈ the snapshot's self-describing manifest, carried by Actor.latest_snapshot_info (12.2).
Actor.Status	the lifecycle (§6.5/§6.6): RUNNING, SUSPENDING/SUSPENDED (external tier), PAUSING/PAUSED (local tier), RESUMING.
LocalSnapshotInfo.node_vms_with_local_snapshots	the warm-node placement index for PAUSED deltas (§10.3); a single source of truth (12.2).
atelet.SandboxAssets / AssetFile.sha256	(node-control-plane material, not

```

ateapi.Actor state)
    the per-architecture sandbox pin (§2.5); the runtime
    binaries content-addressed in the external plane.
ResumeActorRequest.boot
    the skip-golden / boot-from-scratch control selecting
    clean vs golden run mode (§6.6); run_mode persistence
    is the State Root's field.
SuspendActor / PauseActor / ResumeActor (control)
    → Checkpoint / Restore (atelet) → ateom
    drive the write path (§6.4) and resume path (§7);
    imagefsd is the node-local layer beneath them (12.4).

```

12.2 SnapshotInfo migration

Today `ateapi.SnapshotInfo` carries a URI prefix to an opaque snapshot (`ExternalSnapshotInfo.snapshot_uri_prefix`; `LocalSnapshotInfo.snapshot_prefix`), with only `external=2` and `local=3` in the oneof. The locked decision is a backward-compatible oneof extension to a content-addressed, lineage-bearing State Root, not a forklift:

```

// PROPOSED addition to ateapi.proto (state_root = 4 is new):
message SnapshotInfo {
  SnapshotType type = 1;
  oneof data {
    ExternalSnapshotInfo external = 2; // legacy opaque blob (object store)
    LocalSnapshotInfo local = 3; // legacy opaque blob (node VM)
    StateRootSnapshotInfo state_root = 4; // NEW: content-addressed State Root
  }
}
message SnapshotPlacement {
  // warm-node list only; same semantics as
  // LocalSnapshotInfo.node_vms_with_local_snapshots (single source of truth).
  repeated string node_vms_with_local_snapshots = 1;
}
message StateRootSnapshotInfo {
  string state_root_digest = 1; // canonical terrapin-sha256:<hex> of the
  // State Root (§2.1, §6.2)
  SnapshotPlacement placement = 2; // warm-node placement only; tier is CAS state
}

```

- CP-1: A State Root snapshot MUST be referenced through `StateRootSnapshotInfo` carrying the State Root identifier; legacy external/local remain valid for opaque snapshots during migration. The proto additions are a REQUIRED Phase 2 deliverable (§16), landing with hibernate/resume; LLIFS cannot publish/resume a State Root through `Actor.latest_snapshot_info` without them.
 - CP-2: Tier (local/external) MUST be resolved from which CAS/tier holds the State Root's objects (§6.6 TIER-1), consistent with `SnapshotType/CheckpointType`; `SnapshotInfo.type` is informational/ignored for the `state_root` arm and MUST NOT be a second tier authority.
 - CP-3: Lineage (parent) is NOT duplicated in the proto; it lives only in the State Root (§6.2). The control plane MAY cache parent for queries but MUST treat the State Root as authoritative.
 - CP-4: The warm-node placement list MUST have a single representation (`SnapshotPlacement`, reusing the legacy field's semantics); implementations MUST NOT mirror it across competing fields.
 - CP-5: `in_progress_snapshot` semantics MUST be preserved: a State Root MUST NOT be published into `latest_snapshot_info` until the write path (§6.4 WRITE-3) has durably promoted every referenced object, so a crash mid-suspend leaves the prior State Root resumable.
-

12.3 Fork binding

Fork (§6.3) is net-new to the data model: `CreateActor` today derives only from a template. A `create-from-snapshot` needs a `source-snapshot` field:

```

// PROPOSED addition to ateapi.CreateActorRequest:

```

```

string from_state_root = 5; // source State Root id (terrapin-sha256:<hex>);
                             // empty = derive from template. Field number 5 MUST
                             // be reserved for this addition once accepted.

```

CP-6: When `from_state_root` is set, `CreateActor` MUST allocate a NEW actor and the child's genesis State Root MUST set `parent = from_state_root` (§6.3 FORK-1). The source State Root is authoritative for base, memory delta, filesystem delta, runtime delta, sandbox pin, and workload identity; template fields, if supplied, MUST be omitted or match and MUST NOT override inherited state; `worker_selector` applies only as a child placement override.

12.4 Checkpoint/Restore binding

The `atelet/ateom` RPCs today carry only opaque checkpoint references: `atelet Checkpoint/Restore` have a `CheckpointType` plus a local/external configuration oneof; `ateom CheckpointWorkload/RestoreWorkload` carry only an object-storage `snapshot_uri_prefix`. None carry a State Root, delta, or plane fields, so they cannot drive the §6 write / §7 resume path directly. Two conforming options:

CP-7: Either (a) extend these RPCs with State-Root- and delta-capable request/response fields, or (b) front them with a node-local adapter in `imagefsd` that, on `Checkpoint`, ingests the runtime's produced checkpoint into the memory, filesystem, and runtime deltas and emits a State Root (§6.4, DELTA-2R), and on `Restore` resolves a State Root into the page view, the base runtime-state, and the runtime delta the RPC expects (§11.2, RDELTA-5). Until the RPCs carry State Roots, the adapter (b) MUST be provided; neither path may weaken verification (§8) or the freeze-epoch guarantee (§6.4 WRITE-1). These RPC/proto extensions are a REQUIRED Phase 2 deliverable (§16), landing with `hibernate/resume`.

12.5 Run-mode reporting

CP-8: The selected run mode MUST be observable from Actor state/metadata (not the `Actor.Status` enum, which it cannot carry), resolved from the State Root via `Actor.latest_snapshot_info` → the `StateRootSnapshotInfo` `state_root_digest` → the State Root's `run_mode` (§6.2). No separate `Actor.run_mode` field is required; a control plane MAY surface a denormalized copy but the State Root is authoritative.

13. Observability

LLIFS treats latency and runtime-materialization failures as primary observables. The two headline SLOs, TTFE (cold) and TTR (warm), MUST be decomposable into components (§7.6) so a slow start is attributable rather than opaque.

13.1 Requirements

OBS-1: Metrics MUST decompose TTFE and TTR into their components (§7.6) so a start can be attributed as `metadata-`, `proof-`, `network-`, `decompression-`, `verification-`, `restore-`, `copy_up-`, `mmap-`, `policy-`, or `runtime-bound`.
OBS-2: Runtime lazy-materialization failures MUST be reported with clarity equivalent to a classic `ImagePull` failure (§8 FAIL-4); the materialization strictness mode (§7.7) MUST be visible in status, events, and metrics.
OBS-3: A resume that is `restore-bound` (the runtime touched the whole memory image, §7.6 TTR-1) MUST be reported as such, not as a thin lazy resume.
OBS-4: Repeated verification failures and corrupt-source ejections MUST surface as

security events (§14), distinct from ordinary fetch failures.

OBS-5: All tenant-visible telemetry **MUST** be cache-domain-scoped (§9 REDACT-1..3); cross-domain aggregates are operator-only.

13.2 Metrics (recommended)

Latency / SLO:

llifs_time_to_first_exec_ms	llifs_time_to_resume_ms
llifs_mount_latency_ms	llifs_foreground_fault_latency_ms
llifs_memory_fault_latency_ms	llifs_mmap_fault_latency_ms
llifs_time_to_ready_ms	

Verification / proofs:

llifs_terrapin_block_verify_latency_ms	llifs_terrapin_verify_failures_total
llifs_terrapin_proof_cache_hits_total	

Snapshot write / lifecycle:

llifs_snapshot_write_latency_ms	llifs_snapshot_delta_bytes
llifs_reset_per_sec	llifs_wake_latency_ms
llifs_hibernate_delta_bytes	llifs_ram_reclaimed_bytes
llifs_fork_latency_ms	llifs_inplace_rewind_rejected_total
llifs_resident_active_ratio	llifs_per_agent_substrate_overhead_bytes

Restore hazards:

llifs_restore_hazard_refresh_latency_ms	
---	--

Coverage / warmth / drift:

llifs_startup_pack_coverage_ratio	llifs_resume_pack_coverage_ratio
llifs_base_content_coverage_ratio	llifs_memory_base_warmth_ratio
llifs_profile_miss_ratio	llifs_memory_base_overfetch_bytes

Source / hedging / breakers:

llifs_blocks_from_ram_total	llifs_blocks_from_ssd_total
llifs_blocks_from_peer_total	llifs_blocks_from_registry_total
llifs_p0_hedged_requests_total	llifs_p0_hedge_wins_total
llifs_origin_breaker_open_total	llifs_origin_hedge_denied_total
llifs_peer_ejections_total	

Cache / GC / domains:

llifs_cache_evictions_total	llifs_snapshot_write_backpressure_total
llifs_registry_fallback_total	

Copy-up:

llifs_copyup_full_file_fetches_total	llifs_copyup_policy_denied_total
--------------------------------------	----------------------------------

Revocation (§14):

llifs_revocation_state_age_seconds	llifs_revoked_object_denied_total
llifs_revocation_propagation_latency_ms	

Security:

llifs_corrupt_source_ejections_total	
--------------------------------------	--

OBS-6: Metric names are recommended; the decomposition they enable (OBS-1) is required. The following counters **MUST** be present (under any names): per-tier source attribution (so a start attributes to RAM/SSD/peer/registry), origin-hedge denials and origin-breaker openings (§8 ORIGIN-3), and corrupt-source ejections (OBS-4).

13.3 Trace events

image.resolve.{start,done}	rasm.resolve.{start,done}
materialization.assurance.{start,done}	profile.select.{start,done}
bootstrap.fetch.{start,done}	pack.fetch.{start,scatter,verify.done}
mount.ready	container.exec
block.fault	mmap.fault
block.fetch.{start,hedge.start,done}	block.verify.done
origin.breaker.{open,half_open}	hydration.done
snapshot.freeze.{start,done}	snapshot.write.done
resume.start	memory.restore.done
fork.{start,done}	reset.done
hazards.cleared	revocation.denied
	state_root.publish.done
	unfreeze.done
	wake.done
	copyup.detected
	source.ejected (security)

OBS-7: A start/resume **MUST** emit enough trace spans to reconstruct the TTFE/TTR decomposition (OBS-1) end to end.

13.4 Primary SLOs

p50/p95/p99 time-to-first-exec	p50/p95/p99 time-to-resume
p99 foreground fault latency	p99 memory fault latency
p99 mmap fault latency	p99 time-to-ready
startup/resume coverage ratio	proof coverage ratio
base warmth ratio	critical-path network bytes
origin-registry bytes per rollout	runtime lazy-materialization failure rate
reset / wake latency	

OBS-8: p99 TTFE, p99 TTR, and p99 foreground/memory fault latency are the load-bearing SLOs; the substrate SHOULD optimize them before average full-pull or full-hydration time.

14. Security model

The security model is one sentence made enforceable: every byte exposed to a workload is verified against a trusted Terrapin identifier reachable from a trust root, and transport is never a trust anchor. This section names the trust boundary, the requirements that enforce it, revocation, and what is deliberately out of scope.

14.1 Trust boundary

Trusted: the local kernel and container runtime; imagefsd within the node trust boundary; configured trust roots; signed base descriptors and their assurance records (§3); the runtime attestation envelope (§2.5); trusted runtime requirements (§7.2); the revocation policy source (§14.3).

Untrusted or partially trusted: peers; origin-registry transport; mirrors/CDNs; mutable tags; profiles (performance hints only); the network path; unverified local cache entries; compressed bytes before verification.

14.2 Requirements

- SEC-1: Every byte exposed to a workload (filesystem, memory, or proof) MUST be verified against a trusted Terrapin identifier (§2.2 TERRAPIN-4, §5 MVERIFY-1). Peers, mirrors, registries, caches, packs, and decompressed bytes are transport, never trust anchors (§8).
- SEC-2: A Terrapin identifier MUST be reachable from a trust root: a base descriptor validated under the builder regime, or a State Root validated under the runtime regime (§2.5 DIGEST-BIND-4/6). A trusted State Root does NOT confer trust on its base; both MUST be validated (SR-2).
- SEC-3: Profiles MUST NOT alter verification or identity (PROFILE-1); only trusted runtime requirements compel (§7.2).
- SEC-4: The two binding regimes (builder-attested base, runtime-attested per-agent state) MUST NOT be substituted (DIGEST-BIND-5).
- SEC-5: Restore-hazard hooks (§5.6 RHAZARD-1..7) are security-load-bearing for multi-tenant restore from a shared base; a base lacking required hooks MUST NOT be used for multi-tenant restore.
- SEC-6: Cache domains MUST be enforced before peer discovery / Have() and for dedup, keying, warmth, and GC (§9); content-presence side channels MUST be contained within a domain. Cross-domain sharing is opt-in (§9 SHARE-1).
- SEC-7: Peer/source corruption MUST be observable and actionable: a block failing verification MUST eject/penalize the source and MUST emit a security event (§13 OBS-4); repeated verification failure for one object SHOULD trigger

alerting.

- SEC-8: Regardless of isolation, unverified decompressed bytes MUST NEVER reach VFS or guest memory (§8 COMP-3); decompression and untrusted-metadata parsing SHOULD additionally run sandboxed (§8 COMP-4).
- SEC-9: The asserted assurance mode (§3) MUST be reported as a weaker mode and MUST NOT claim node-verified OCI+rootfs equivalence.
- SEC-10: A mutable tag MUST NOT be treated as a trust root: it MUST be resolved to an immutable OCI image digest and a validated base descriptor before any byte is exposed (§8 REG-4, §7 PROFILE-3). A gap between tag discovery and digest binding MUST NOT widen trust.

14.3 Revocation

- REV-1: Signatures and attestations (base descriptors, materialization attestations, runtime attestations) SHOULD have short validity windows unless backed by an online revocation check.
- REV-2: The substrate SHOULD support revocation sources (transparency-log inclusion policy, CRL/OCSP-like checks, or signed deny lists); revocation state MUST be cached with an expiry.
- REV-3: If a base descriptor (or its committed OCI image digest, §2.5 DIGEST-BIND-3), profile record, runtime-requirements policy, or State Root is revoked, the control plane MUST stop new scheduling/resume against it. Revocation evaluation MUST include both lineage enumeration (§6.3 LINEAGE-2) and State Root `base`-reference resolution: revoking a base MUST deny every retained State Root whose `base` reference (§6.2) resolves to that base descriptor, and thus its OCI image digest. (State Roots do not themselves carry an OCI subject; the chain is State Root.base → base descriptor → OCI digest.)
- REV-4: Policy MUST define whether already-running agents bound to a revoked base are allowed to continue, drained, killed, or isolated.
- REV-5: Revocation propagation latency SHOULD be observable (§13).

14.4 Deliberately out of scope (deployment must supply)

This document specifies verified state delivery; it does NOT by itself provide workload authentication/authorization or network egress policy. Until those are added, a deployment relying on LLIFS alone for isolation MUST run trusted workloads in an isolated cluster.

- SCOPE-1: Workload authn/authz and network/egress policy (e.g. NetworkPolicy) are NOT provided by LLIFS and MUST be supplied by the surrounding platform for untrusted multi-tenant workloads.
- SCOPE-2: High-availability/multi-replica of the control plane, and full control-plane sharding, are out of scope here; the cache-domain keying and content-addressed identity (§9) and the byte-exact encodings (§15) are fixed so they can be added without changing identity.
- SCOPE-3: At-rest encryption of private memory/filesystem/runtime deltas (§6) SHOULD be applied in the cold/blob tier for confidential deployments; encryption disables cross-tenant dedup of those objects (which §9 already forbids by default), so it composes with the per-domain keying without weakening it.

15. Canonical encodings and conformance vectors

Every identity in LLIFS is a Terrapin digest over a byte-exact canonical encoding, so independent implementations agree or reject (the supply-chain equivalence claim, MAT-20). This section is the registry of those encodings, pointing to the two already defined inline, defining the remaining (agent-state) ones, and pinning the conformance vectors that make them testable.

15.1 Shared primitives

All canonical encodings in this document use the §3.2 primitives (PRIM-1..5): u8/u16/u32/u64 unsigned little-endian; i64 two's-complement little-endian; varbytes = u32 length prefix + raw bytes; a 32-byte digest field is the raw Terrapin identifier (the G(manifest) payload, never the hex string, never the bare tree root). Unless stated otherwise, every canonical document is length-framed and a parser MUST reject trailing bytes.

15.2 Encoding registry

filesystem logical tree	LLT1	§3.2	→ logicalRootfsDigest
filesystem physical layout	LLAY1	§4.3	→ layoutDigest
State Root	LLSR1	§15.4	→ State Root identifier
base descriptor	LLBD1	§15.5	→ base descriptor identifier
memory layout	LLML1	§15.5	→ memory layout identifier
memory delta (page-indexed)	LLMD1	§15.6	→ memory delta identifier
filesystem (overlay) delta	LLFD1	§15.7	→ filesystem delta identifier
runtime delta (opaque)	LLRD1	§15.4	→ runtime delta identifier
workload identity	LLWI1	§15.8	→ workload_identity
sandbox pin	LLSP1	§15.9	→ sandbox pin (external sha256)

ENC-1: A new object's identity MUST be the Terrapin identifier of its canonical encoding here (the sandbox pin is the exception: an external-plane sha256, §15.9). LLT1/LLAY1 (§3.2/§4.3) are normative as written and not restated.

ENC-2: Every canonical document MUST begin with its 4-byte magic and a u16 version, MUST length-frame its records, and MUST reject trailing bytes and unknown enum values.

15.3 Terrapin profile conformance vectors

Computed by the reference oracle (real values, not illustrative). These constants are load-bearing and PROVISIONAL until independently reproduced by a second implementation/oracle (ENC-CONF-2); they MUST be cross-checked before §15 is frozen for release.

Constants:

G(empty) = 473a0f4c3be8a93681a267e3b1e9a7dcda1185436fe141f7749120a303721813
 G(2 MiB zero block) = 67cbcd9b97ddabde2863f4daefa4f57176567a7c3ccfa1560c1065f9c8af74d6

Identifier vectors (terrapin-sha256:<hex> = G(canonical manifest)):

empty (len 0) = f4b8abc1cfd6ffec75b4070be5440706286b3a7af937ef5d020ca2c0c1210458
 "hello world" (len 11) = 7bc0163f32e5f6082308ae0dff3dc7c9b0488e5aa652d9de01418df5ec800c8c
 one zero byte (len 1) = dce39f984d9c140e4ad8f4b448a2ae6ae5398ed1adbb4d07ed8bedbc5b3b4598
 block-1 (2097151 zero) = dc7f0a33cf02e7a84fc380a41d396b451c96325a633a87528ebf797621befad7
 one block (2097152 zero) = 6fbd6447c2d8d70a83ae159461847a1a410679900702433dd2b04d063a3b2f9b
 block+1 (2097153 zero) = 5ba8049ae8f68a47acd4fad265c8a963aa82735e90f209dd79ff8d6d2188fdc5
 128 GiB zero (65536 blk) = 8d03319328c6d6b3cd00566d894443b2a82d31437b580ee533c2021d82bdb5a4
 128 GiB + 1 byte = 6f552f944f4995878c7facc92c29c3643aaafc2a5bff90e255bbf430210d551b

Manifest accept/reject (canonical Terrapin manifest, §2.2): a manifest MUST be ASCII, LF-terminated (including the last line), field order exactly terrapin, block_size, length, tree, exactly one space after each colon, integers with no leading zeros, tree exactly 64 lowercase hex. Each of these MUST be rejected (not normalized): uppercase hex; missing final LF; wrong field order; leading zeros; extra spaces; unknown/extra keys, comments, or blank lines; block_size != 2097152; the bare tree root presented as the identifier.

15.4 State Root encoding (LLSR1)

magic "LSR1"; version u16=1; then fields in fixed order; varbytes fields are

length-framed (so "none" is an explicit length=0 value and optional digests are length 0 or 32, never a mix), and the createdSec/createdNsec fields use their stated fixed widths:

```
actor (varbytes); parent (0 or 32 = State Root id); base (32 = base descriptor
id); memory (0 or 32 = memory delta id); filesystem (0 or 32 = filesystem delta
id); runtime (0 or 32 = runtime delta id); sandbox_pin (0 or 32 = sha256);
workload (0 or 32 = workload_identity);
run_mode (varbytes of one byte: 1=clean 2=golden 3=persist-rootfs
4=persist-rootfs+memory); producer (varbytes); createdSec (8-byte i64);
createdNsec (4-byte u32, 0 <= nsec < 1e9).
```

State Root identifier = Terrapin identifier of these bytes.

ENC-SR-1: All fields MUST be present in this order; "none" is the explicit length=0 value, never omitted (OBJ-5, SR-1). A digest field's length MUST be exactly 0 or 32; any other length MUST reject. Unknown run_mode MUST reject. createdNsec >= 1e9 MUST reject.

ENC-SR-2: Exactly the §6.2/SR-4 conditional rules apply: if memory or filesystem is non-none, workload MUST be non-none; if memory is non-none, sandbox_pin MUST be non-none. When memory is non-none, runtime is normally non-none and is "none" only under the RDELTA-1 no-runtime-state policy exception; restore enforces RDELTA-5.

Runtime delta encoding (LLRD1), the State Root's runtime field (§6.1 DELTA-2R):

```
magic "LRD1"; version u16=1; payload (varbytes, opaque runtime-native bytes
captured at freeze). Runtime delta identifier = Terrapin identifier of these bytes.
```

ENC-RD-1: payload is OPAQUE to LLIFS (no internal structure, like the base runtime_state blob, §15.5 ENC-BD-3); its Terrapin identifier covers the exact stored bytes; trailing bytes MUST reject. A later version MAY define a structured delta against the base runtime_state blob.

ENC-RD-2: A restorer MUST interpret a runtime delta ONLY under a runtime satisfying the State Root's sandbox pin (§2.5) and the base descriptor's compatibility matrix (§11.4), and MUST apply it only after the base runtime-state blob (RDELTA-5); an incompatible or missing-required runtime delta MUST fail closed (RDELTA-1/5).

15.5 Base descriptor encoding (LLBD1)

```
magic "LBD1"; version u16=1; then fields in this exact order; varbytes fields
length-framed, fixed-width fields (32-byte digests, u8, u32) at the stated width:
oci_image_digest (varbytes: the ASCII string "sha256:<64 lowercase hex>",
ENC-BD-4); logical_rootfs (32); layout (32);
base_memory (32); runtime_state (32); memory_layout (32); sandbox_pin (32
sha256); compat_matrix (the CompatMatrix sub-record below); assurance_mode
(u8: 1=asserted 2=derived-attested 3=node-derived).
```

```
CompatMatrix (fixed field order, each as typed): runtime_id (varbytes);
runtime_version (varbytes); platform_mode (u8: 1=ptrace 2=systrap 3=KVM);
arch (varbytes, GOARCH); page_size u32; cpu_feature_mask (varbytes, opaque
fixed-width feature bitset bytes); checkpoint_format_version u32; abi_tag
(varbytes); seccomp_tag (varbytes); cgroup_tag (varbytes); namespace_tag
(varbytes).
```

Base descriptor identifier = Terrapin identifier of these bytes (the signed subject, §2.5 DIGEST-BIND-3).

ENC-BD-1: All fields and CompatMatrix sub-fields MUST appear in this exact order and width; unknown platform_mode or assurance_mode MUST reject; trailing bytes MUST reject. No tag is sorted (each is a single value, not a list); all are length-framed so empty is length=0, distinct and intentional. cpu_feature_mask is OPAQUE producer-captured bytes committed verbatim: its width and bit assignment are arch-defined and are NOT independently re-derived for byte-identity (the descriptor commits exactly the bytes the producer captured). A restorer compares its own capabilities against the mask using the arch's interpretation (§11.4 MATRIX), but the committed bytes are authoritative for identity.

ENC-BD-2: The base descriptor is the signed trust subject; a verifier MUST reject a descriptor that does not parse to exactly this canonical form.

ENC-BD-3: runtime_state is a runtime-native OPAQUE byte blob: LLIFS defines no

internal structure for it, and its Terrapin identifier covers its exact stored bytes verbatim. (It is runtime-version-specific; compatibility is gated by the compatibility matrix, §11.4.)

- ENC-BD-4: `oci_image_digest` MUST be the canonical OCI digest STRING form, the exact ASCII bytes "sha256:" followed by exactly 64 lowercase hex characters, and nothing else (it is the external-plane identity, §2.1; raw 32-byte form, uppercase hex, other algorithms, or a missing prefix MUST reject). It is the only base-descriptor field in the external plane; all other digest fields are raw 32-byte Terrapin identifiers.

Memory layout encoding (LLML1), the base descriptor's `memory_layout` field (§5.3):

magic "LML1"; version u16=1; page_size u32; regionCount u64; regions (sorted by guestStart, non-overlapping), each: guestStart u64; length u64; state u8 (1=mapped-data 2=mapped-zero 3=unmapped/no-access, §5.3 MLAYOUT-5); objectOffset u64 (offset into the base memory snapshot object for mapped-data; 0 otherwise); prot u8 (bit0 R, bit1 W, bit2 X); flags u8 (bit0 guard-page, bit1 shared else private).

Memory layout identifier = Terrapin identifier of these bytes.

- ENC-ML-1: regions MUST be sorted by guestStart, non-overlapping, and tile the committed guest address space [0, committedEnd) exactly (committedEnd = guestStart+length of the last region); unknown state/prot/flags bits MUST reject; trailing bytes MUST reject. This is the byte-exact form of the §5.3 MLAYOUT-1.5 mapping (linear order, three-state mapping, page size, protections); it commits the per-range mapping state and attributes that restore reproduces.
- ENC-ML-2: The base memory snapshot is guest-address-linear (MLAYOUT-1): for a mapped-data region (state 1) objectOffset MUST equal guestStart; for mapped-zero (2) and unmapped/no-access (3) objectOffset MUST be 0 and the region occupies no stored bytes, contributing canonical zeros to the base-memory identity (MEM-1). The base memory snapshot object's manifest length MUST equal committedEnd. Because the MEM-1 identity does not distinguish states 2 and 3 (both zeros), this signed layout is the sole authority for that distinction (MEM-5); any other objectOffset MUST reject.

15.6 Memory delta encoding (LLMD1)

The memory delta is a page MAP: for each diverged guest page, where its bytes come from (§6.1). The map directly names every page's source, so resume needs no ancestor traversal (§6.3 LINEAGE-3) and a fork shares parent/base pages by direct reference with no copy.

magic "LMD1"; version u16=1; page_size u32; base_ref (32, the base memory snapshot object); entryCount u64; entries, each:
 guestPageIndex u64;
 source u8 (1=this-delta-data 2=ancestor-delta-data 3=base 4=canonical-zero);
 sourceObjectId 32 (the data object for source 1/2; all-zero for source 3/4);
 offset u64; length u64
 sorted ascending by guestPageIndex, unique. The delta's own changed pages live in a separate "delta-data" Terrapin object referenced by source==1 entries.
 Memory delta identifier = Terrapin identifier of these (map) bytes.

- ENC-MD-1: The encoding MUST be canonical (one form per restored memory state). entries MUST be sorted and unique by guestPageIndex. A page identical to the base MUST be OMITTED (it resolves from the base copy-on-write, §5.4); source==3 (base) is used ONLY by a fork to explicitly revert a page the parent had changed back to the base value, never to redundantly list a base-identical page. For a fork, unchanged-from-ancestor pages use source==2 naming, by absolute Terrapin id, the ANCESTOR delta-data object that actually holds the bytes (resolved ONCE at fork time by reading the parent's map, the §6.3 FORK-3 flatten), so the child map is self-contained and resume never consults any ancestor map; changed pages use source==1. Unknown source MUST reject; trailing bytes MUST reject.
- ENC-MD-2: Each entry covers exactly one guest page: length MUST equal page_size. For source==1/2, [offset, offset+length) MUST be in-bounds of the named data object; source==2's sourceObjectId MUST be the absolute Terrapin id of the ancestor delta-data object holding the bytes, which resume reads

directly and MUST NOT resolve through any ancestor's page map (LINEAGE-3). For source==3 (base), sourceObjectId MUST be all-zero and offset MUST be the page's guest offset within the base memory snapshot. For source==4 (canonical zero), sourceObjectId MUST be all-zero and offset MUST be 0. On restore each page is resolved from its named source and verified (§5 MVERIFY) before exposure (source==4 installs a verified zero page, §5.4 COW-6b); every source object is a separate Terrapin object.

- ENC-MD-3: The delta's own changed pages MUST form exactly ONE canonical "this-delta-data" Terrapin object: the source==1 pages concatenated in ascending guestPageIndex order. For each source==1 entry, sourceObjectId MUST equal that one object's Terrapin id, length MUST equal page_size, and offset MUST equal (the page's ordinal among source==1 entries) * page_size. This makes LLMD1 canonical: the same changed-page state over the same base/parent yields one memory delta identifier.
- ENC-MD-4: A memory delta MUST be directly resolvable (DELTA-5): every entry names, by absolute Terrapin id, the exact object holding its bytes (this-delta-data, an ancestor delta-data object, or the base via base_ref), and resume reads those named objects directly, never walking lineage or consulting any parent delta's map. base_ref and the source==2 ancestor object id are DIRECT resolution inputs; only the parent State Root link and lineage chain (§6.3) are provenance, GC, and compaction inputs.
- ENC-MD-5: Source selection MUST be canonical (one encoding per effective page state), in this precedence: (a) OMIT a page byte-identical to the base (it resolves from base copy-on-write, §5.4); (b) use source==3 (base) ONLY to revert a fork page the parent had changed back to the base value, never for an already-base-identical page (omitted per (a)); (c) use source==4 (canonical zero) ONLY when the page is all-zero AND the base page at that offset is not all-zero (an all-zero page equal to the base is omitted per (a)); (d) otherwise use source==1 for a changed page and source==2 for a page carried unchanged from an ancestor. A verifier MUST reject structurally detectable violations (ENC-MD-2) and, where it has the base, a redundant source==3/4 for a base-identical page.

15.7 Filesystem (overlay) delta encoding (LLFD1)

The canonical overlay-delta object reconstructing the writable upper, §6.1. It is a FINAL-STATE delta: it encodes the resulting writable upper directly, not a replay log, so there are no multi-step operations (a rename is encoded as its result: a put at the new path and a whiteout at the old). It is SELF-CONTAINED against the base lower: a resume reconstructs the upper from this object alone, plus the content objects it names by absolute id and the base rootfs the State Root references, never by walking parent deltas (DELTA-5, LINEAGE-3). Op tails reuse the LLT1 common prefix exactly.

```
magic "LFD1"; version u16=1; parent_ref (varbytes: 32-byte filesystem delta id, or
length-0 = none; PROVENANCE/GC/compaction only, never required to reconstruct the
upper, DELTA-5); opCount u64; ops (sorted by normalized path bytes), each an
OpRecord:
  path (varbytes, normalized per §3.2);
  op u8 (1=put-file 2=put-dir 3=put-symlink 4=put-chardev 5=put-blockdev
        6=put-fifo 7=whiteout);
  then the op tail:
    put-* common prefix (exactly LLT1 §3.2 widths, in order): mode u32; uid u32;
      gid u32; mtimeSec i64; mtimeNsec u32; xattrCount u32; xattrs (XattrRecords
      sorted by key, unique; §3.2);
    then the type tail: put-file → contentId 32 + contentLen u64; put-symlink →
      target varbytes; put-chardev/put-blockdev → major u32 + minor u32;
      put-dir/put-fifo → (none);
      whiteout → (no tail).
```

- ENC-FD-1: ops MUST be sorted by normalized path and paths MUST be unique (exactly one op per path; a path cannot be both put and whiteout), and all paths MUST normalize per §3.2 (reject absolute/NUL/".."). Because the delta is final-state, there is no rename op and no multi-step conflict to resolve. hardlinkGroup is NOT represented: an overlay copies up whole files, so a put-file carries its own contentId (a later compaction MAY dedupe by content object, but the delta encodes no hardlink group). Unknown op MUST reject; trailing bytes MUST reject.

- ENC-FD-2: whiteouts MUST be representable so deletions survive restore. The filesystem delta identifier is the Terrapin identifier of these bytes; content objects are separate.
- ENC-FD-3: The filesystem delta MUST be directly resolvable (DELTA-5): it encodes the COMPLETE final-state writable upper and every content reference is an absolute content-object id, so resume reconstructs the upper from this object, its content objects, and the base lower alone. parent_ref is the lineage edge for provenance/GC/compaction and MUST NOT be consulted on the resume critical path; a compaction MAY rewrite a parent-relative storage form into this complete form, but resume MUST NOT depend on traversal.

15.8 Workload identity encoding (LLWI1)

The §6.2 SR-5 canonical workload identity (an internal Terrapin identity, not an external digest).

magic "LWI1"; version u16=2; then a pod block, then containers.

Pod block: pause_image (varbytes); hostname (varbytes); host_flags u8 (bit0 hostNetwork, bit1 hostPID, bit2 hostIPC, bit3 shareProcessNamespace); pod security (SecurityContext sub-record below).

containerCount u64; containers (in declared order), each:

- name (varbytes); image (varbytes);
- cmdCount u64 + each (varbytes, declared order);
- argsCount u64 + each (varbytes, declared order);
- envCount u64 + each (name varbytes; value varbytes), SORTED by name then value;
- working_dir (varbytes);
- security (SecurityContext sub-record);
- capAddCount u64 + caps (varbytes, SORTED, unique);
- capDropCount u64 + caps (varbytes, SORTED, unique);
- mountCount u64 + mounts (SORTED by mountPath), each: mountPath (varbytes);
- subPath (varbytes); sourceName (varbytes); sourceType (varbytes); mflags u8 (bit0 readOnly);
- deviceCount u64 + devices (SORTED by path), each: path (varbytes);
- mem_limit u64 (bytes, 0 = unset); cpu_limit_milli u64 (0 = unset).

SecurityContext sub-record: runAsUser i64 (-1 = unset); runAsGroup i64 (-1 = unset); sflags u8 (bit0 privileged, bit1 readOnlyRootFilesystem, bit2 allowPrivilegeEscalation, bit3 runAsNonRoot); seccomp (varbytes); apparmor (varbytes); selinux (varbytes).

workload_identity = Terrapin identifier of these bytes.

- ENC-WI-1: Fields are the restore-visible projection of atele.WorkloadSpec (which carries them); atele's reduced Container is insufficient (§12). Every list is ALWAYS present with its count; absent and empty lists are identical (count 0). env, capAdd, capDrop, mounts, and devices MUST be sorted as stated and unique by their sort key (a duplicate env name, capability, mount path, or device path MUST reject). Unset scalars use their stated sentinel (i64 -1, u64 0, varbytes length 0). The encoding MUST be deterministic. (This is the value of the State Root `workload` field, §6.2; "workload_spec_digest" is a deprecated alias for workload_identity.) The exact field set MUST be reconciled against atele.WorkloadSpec before §15 is frozen. This is a HARD freeze blocker, not a nicety: a workload-identity mismatch is a restore-safety bug, so any WorkloadSpec field that affects restore correctness MUST be added here (SR-5) before v1.0 freezes.
- ENC-WI-2: Fields intentionally EXCLUDED, because they do not affect restored state or are regenerated post-restore, MUST NOT be committed: scheduling and placement (node name, affinity, tolerations, priority, topology); status and conditions; liveness/readiness/startup probes; restart policy; image pull policy and pull secrets (provenance, not restore state); purely-informational labels and annotations; dynamically assigned values (pod IP, assigned ports, service-account token contents) and other RHAZARD-refreshed state (§5.6); and volume CONTENTS (the data lives in the filesystem plane; only mount TOPOLOGY is committed, above). An annotation that materially changes runtime execution is NOT informational and MUST be committed (SR-5).

15.9 Sandbox pin encoding (LLSP1)

Per-architecture, external-plane identity of the runtime assets (§2.5).

```
magic "LSP1"; version u16=1; sandbox_class (varbytes); goarch (varbytes);
assetCount u64; assets (SORTED by name bytes), each: name (varbytes),
asset_sha256 (32). AssetFile.url is excluded (transport hint, not identity).
sandbox pin = sha256 over these bytes (external plane, §2.1).
```

ENC-SP-1: The pin covers exactly one GOARCH; other architectures' assets are not included (§2.5 DIGEST-BIND-7). sandbox_class and goarch MUST be non-empty; the asset list MUST be non-empty; assets MUST be sorted by name and unique; trailing bytes MUST reject.

15.10 Conformance and the reference oracle

ENC-CONF-1: A conforming implementation MUST reproduce the §15.3 vectors and the LLT1/LLAY1 golden vectors, and MUST agree byte-for-byte on every canonical encoding here or reject identically.

ENC-CONF-2: A reference oracle (a non-production tool sharing the Terrapin core) SHOULD emit the vector set as the cross-implementation conformance target, covering: the §15.3 Terrapin vectors; LLT1 cases (sorted entries, xattrs, hardlink pair, sparse vs dense same identity, zero-length file, and the reject cases: absolute/NUL/".." paths, duplicate path, bad type, duplicate xattr, mtimeNsec overflow, count mismatch, trailing bytes); LLAY1 cases (tiling extents, virtual-zero, empty and hardlink files, plus the reject cases including backing==2 / non-zero inlineCount, which are reserved in v1, LLAY1-11); a memory-layout (LLML1) case with a mapped-data region at a HIGH guest address preceded by a very large unmapped/no-access zero span (to catch tree-height and off-by-one bugs in the virtual zero-subtree proof rules, SPARSE-8), plus a mapped-zero vs unmapped state-mismatch reject case; an LLSR1 round trip WITH a non-none runtime delta (LLRD1) present; and at least one round trip for LLSR1/LLBD1/LLMD1/LLFD1/LLRD1/LLWI1/LLSP1 plus their reject cases.

ENC-CONF-3: The production digest-producing code SHOULD be a single implementation (the digest boundary is the language boundary) so independent implementations cannot silently diverge; the oracle's vectors are the agreement test.

16. Phase plan and minimal viable surface

The core bet: injecting exactly the right verified bytes at startup/resume beats making the pipe wider, and density comes from ONE verified base shared copy-on-write across many sandboxes. The first gate is therefore a single capability, not the whole lifecycle surface: shared-copy-on-write base memory in gVisor (§11.2 GVISOR-3). Phase 1 proves exactly that gate on one node (nothing more); Phase 2 builds the persisted lifecycle and production latency on top; Phase 3 adds yield, fan-out, and fleet behavior; Phase 4 optimizes with native backends. Each phase has a success criterion.

16.1 Phase 1: prove the density gate (one node)

Phase 1 is NOT the full lifecycle product; it is the shortest path to proving the density mechanism, built as a dependency ladder:

```
Rung 1 – Verified lazy filesystem. Terrapin verification (§2.2) + LLT1/LLAY1
(§3, §4) served through a CAS-backed gofer (gVisor) or FUSE lower with
```


verify-before-expose lazy reads, singleflight, P0 priority, hedged fetch + origin breaker, registry/mirror fallback (§8); OverlayFS upper + copy-up (§4.5). A conformance oracle emits the §15 vectors (§15.10). Gate: a multi-GiB image execs without full hydration; every exposed byte verifies.

Rung 2 – Lazy restore page delivery. Reuse gVisor background restore (§11, GVISOR-2) with the page source redirected into the CAS-backed gofer/page provider: a base memory snapshot (§5.1, §5.3) faults lazily, verify-before-expose, with the absent / known-zero / unmapped hazard handling (§5.4 COW-6/6a/6b) and the restore working set (§5.5). Pages are still materialized per sandbox here; density arrives at rung 3. Gate: a sandbox restores and runs faulting only what it touches, and an absent range never reads as zero.

Rung 3 – Shared copy-on-write base (THE DENSITY GATE). The one net-new runtime capability (§11.2 GVISOR-3): N sandboxes MAP_PRIVATE one verified base file so resident base pages are physically shared and only written pages go private. Until this passes, everything above is a latency optimization, not a density breakthrough. Gate: the acceptance test below passes.

Do NOT build P2P, fork, yield, or the full delta/State-Root lifecycle in Phase 1. The control-plane binding (§12) lands in Phase 2 with hibernate/resume.

Acceptance test (the density gate, deliberately brutal). Restore N sandboxes from one base and prove:

- a base block is FETCHED once and VERIFIED once, independent of N (verification cost does not scale with agent or fault count, MVERIFY-2);
- the same host PHYSICAL pages back all N sandboxes (one resident copy per cache domain, COW-1/COW-3);
- a write creates a private page for the writer ONLY (copy-on-write, no bleed);
- an absent (not-yet-fetched) range NEVER reads as zero: it traps, fetches, and verifies before exposure (COW-6/6a);
- a signed known-zero range installs a verified zero page with NO fetch (COW-6b), distinguished from absent by the signed layout (MEM-5), never by a memfd hole.

If this passes, the density thesis is real; if not, the rest is a lazy checkpoint reader, not a density substrate.

Measurement harness (the first prototype, before any platform features). Not a control plane: a harness around the shared base. Launch a multi-GiB warmed gVisor workload, capture a base, restore many sandboxes, and report base resident bytes, proportional set size (PSS) per sandbox, dirty pages per sandbox, Terrapin verification count, foreground memory-fault p50/p95/p99, absent-range fault behavior, known-zero behavior, and TTR (§7.6). These numbers, not a feature list, are what distinguish a real density substrate from a merely lazy restore.

16.2 Phase 2: persisted lifecycle and production latency

Build the persisted lifecycle on the proven base, then harden latency.

Lifecycle: overlay-delta capture (§6.1: memory, filesystem, and runtime deltas) with self-contained deltas (DELTA-5); the write/commit path under the single freeze barrier (§6.4); State Root mint + hibernate/resume (§6.2, §6.5) with local/external tiers and run modes (§6.6); then copy-on-write FORK (§6.3) with the authorization hook (FORK-5). Control plane: the ateapi SnapshotInfo → StateRootSnapshotInfo migration, SnapshotPlacement, CreateActorRequest.from_state_root, and the Checkpoint/Restore State-Root binding (§12) are REQUIRED here.

Latency hardening: RAM CAS; an eBPF/VFS startup+resume profiler and dynamic profile refinement; scheduler warmth scoring + resume affinity (§10); controller-driven prewarm; registry metadata cache; derived-attested as the default assurance mode (§3).

Success: a hibernated agent resumes from its delta + shared base with no full hydration and every byte verified; a fork fans one snapshot into N children; and p99 foreground-fault and memory-fault latency stay bounded across realistic rollouts while profile drift is detected and re-profiled.

16.3 Phase 3: yield, fan-out, and fleet

Add yield and scale out.

Yield: hibernate-under-upstream-call plus the durable upstream transaction (§6.5 YIELD-1..3) with at-most-once submission and the sent_unknown state machine (YIELD-2/2a).

Fleet: the peer protocol (§8.6); rack/AZ-aware source selection and seed-node rollout planning; peer health scoring; cross-domain peer policy; cluster-wide origin-hedge budget coordination and origin breakers at scale; fork fan-out at scale (§10.4). Do NOT build P2P before this phase.

Success: a blocked agent yields its RAM and wakes to deliver the held response with no duplicate non-idempotent upstream execution beyond declared policy; and origin-registry bytes per 1,000-node rollout drop substantially while TTFE/TTR stay within SL0, a fork of one agent into N children fanning out from peers, not origin.

16.4 Phase 4: native performance path

Evaluate native, page-cache-integrated backends (optimized FUSE, EROFS/fscache, composefs-style verified trees, virtiofs) per §11.5, and the in-place-rewind reset optimization (§6.5 RESET-4) only if measured to beat restore-fresh.

Success: a native backend improves p99 fault latency, CPU, and memory footprint without weakening logicalRootfsDigest, layoutDigest, Terrapin, State Root, or cache-domain guarantees.

16.5 Minimal viable surface

The minimal viable surface is the complete v1 component set, in dependency order below. Phase 1 (§16.1) builds only the density-gate subset: items (1)–(4) plus the shared-CoW base of (10); the persisted lifecycle of (5) lands in Phase 2 and the fleet items (8)–(9) and yield in Phases 2–3. (1) the two digest planes + Terrapin verification (§2); (2) materialization + LLT1 (§3); (3) content-addressed layout + LLAY1 (§4); (4) the base memory snapshot + CoW restore + RHAZARD (§5); (5) overlay/delta + State Root + lineage + fork + reset/hibernate/yield + tiers/run-modes (§6); (6) profiles/requirements/packs/bootstrap/coverage + TTFE/TTR (§7); (7) the verify-before-expose fetch path + singleflight + hedging + origin breaker + registry fallback (§8); (8) cache domains + keying + leases + mark-sweep GC (§9); (9) warmth + resume affinity (§10); (10) the gVisor adapter incl. shared-CoW base (§11); (11) the atepi/atelet/ateom binding incl. the proposed proto additions (§12); (12) the canonical encodings + conformance oracle (§15); (13) baseline observability (§13) and the security/revocation model (§14). P2P (§8.6, Phase 3) and native kernel backends (§11.5, Phase 4) come later.

16.6 Deliberately deferred

Per §14.4: workload authn/authz and network/egress policy (SCOPE-1); control-plane HA and full sharding (SCOPE-2); at-rest encryption of private deltas (SCOPE-3). A deployment relying on LLIFS alone for isolation MUST run trusted workloads in an isolated cluster until SCOPE-1 is supplied.

End of specification.

17. Glossary

Canonical terms (the only terms this document uses for stored state and lifecycle):

block	2 MiB Terrapin fetch+verify unit. (Was: "chunk".)
content object	Terrapin object = one file's content bytes; filesystem dedup unit. (Was: "artifact"; "chunk map" for files.)
base rootfs	Shared read-only filesystem base (LLT1 tree + content objects via LLAY1).
base memory snapshot	Shared read-only 2 MiB-linear guest-memory image. (Was: "golden memory" as a distinct object; "golden" is now an informal adjective only.)
base descriptor	Signed identity binding OCI image digest + base rootfs + base memory + runtime-state + memory layout + sandbox pin + assurance mode, as one unit (§2.5, DIGEST-BIND-3). (Subsumes the earlier "checkpoint manifest".)
overlay	Live per-agent writable layer (dirty pages + fs upper); ephemeral, never in shared CAS. (Was: "divergence".)
delta	Persisted content-addressed snapshot of an overlay, in three parts: memory delta = a page MAP naming each diverged page's source (this-delta/ancestor/base/zero, §15 LLMD1); filesystem delta = overlay-delta object (content-object refs + namespace/metadata ops + whiteouts, §15 LLFD1); runtime delta = opaque runtime-native non-memory state (§15 LLRD1). (Was: "plane manifest"; "chunk map" for memory.)
State Root	Content-addressed identity of a persisted agent snapshot, binding actor + base ref + memory delta + filesystem delta + runtime delta + sandbox pin + workload identity + run mode + producer + created + parent.
actor	The stable identity of an agent; survives across all of its snapshots' State Roots. A fork allocates a NEW actor (§6.3).
lineage	Parent-pointer DAG over State Roots; basis of forking.
reset / hibernate / yield	RAM-reclamation mechanisms (§6): reset = restore base with no delta; hibernate = base + stored delta; yield = hibernate during a blocking upstream call.
tier (local/external)	Where a delta lives: local (node SSD) or external (object storage). Orthogonal to the reclamation mechanism; maps to the control plane's PAUSED (local) / SUSPENDED (external).
cache domain	The maximum scope within which content-presence timing leakage is acceptable; the isolation unit for dedup, keying, warmth, and GC (§9).
warmth / coverage	The fraction of an object/profile's required blocks (or content objects, or base) already present on a node (§10).
TTFE / TTR	Time-to-first-exec (cold start) / time-to-resume (warm snapshot restore) (§7).
profile	A weakly-trusted hint recording the working set faulted earliest (startup profile = cold, resume profile = warm); optimizes, never compels (§7.1). Keyed per PROFILE-3.
runtime requirements	Trusted policy that COMPELS (mandatory prehydration, fail-closed, expected-mutable handling); from signed metadata/admission/operator config, never an untrusted profile (§7.2).
pack	A transport envelope of canonical BlockRefs (startup pack = cold, resume pack = warm), scatter-extracted into the CAS; carries no Terrapin identity of its own (§7.3).
bootstrap	The compact metadata record carrying all fetch + trust inputs so a start needs no metadata/proof/attestation round trip (startup/resume bootstrap, §7.4).
coverage	The fraction of a required set (BlockRefs, or content objects/base for warmth) already present on a node (§7.5, §10).
singleflight	The per-BlockRef state machine that collapses concurrent reads/faults for one block into a single in-flight fetch (§8.2).
hedge	A redundant fetch issued for a slow P0 request; first VERIFIED response wins, losers cancel (§8.4).
origin / mirror / peer / source	Fetch sources: origin = the authoritative registry; mirror = a near-cluster pull-through cache/CDN; peer = another node (untrusted for integrity, §8.6); source = any of these chosen by source selection (§8.5).
origin breaker	A per-registry/per-image circuit breaker + budget that stops P0 hedges from stampeding a degraded origin (§8.4).
lease	A reference that pins an object alive in a cache domain;

	types include runtime/startup/profile/peer/operator-pin/cache-domain/materialization/pause/state-root (§9.2). Leases are GC roots (§9.3).
template/operator lease	A lease pinning shared base objects while a template is live, so the base is not GC'd by per-agent churn (§9.3).
epoch	The version of a per-domain cache key; rotation bumps the epoch and writes under the new key without invalidating existing entries (§9.1).
trusted-shared / private-domain	Cache keying modes: trusted-shared keys by the plain BlockRef (cross-domain dedup, opt-in); private-domain keys by HMAC(domainKey[epoch], BlockRef) so presence does not leak across domains (§9.1).
mark-sweep	The GC model: retain every object reachable from a root (live lease or live State Root), sweep the rest (§9.3).
logicalRootfsDigest	Terrapin digest of the canonical logical rootfs tree (LLT1, §3); the supply-chain equivalence claim.
layoutDigest	Terrapin digest of the canonical physical layout (LLAY1, §4).

Supporting terms (structural objects, committed fields, and infrastructure, not agent-state object classes, see §2.3):

page	The platform MMU page (4 KiB on x86-64); the unit of per-agent copy-on-write memory sharing (§2.4, §5).
CAS / node CAS	The node-local content-addressed store of verified blocks, keyed per §9.
imagefsd	The privileged node-local LLIFS daemon: resolves metadata, maintains the CAS, fetches and verifies blocks, serves reads (§8, §11).
runtime-state blob	The non-memory sandbox state needed to resume (threads, fds, namespaces, vCPU/sentry state); a structural object committed by the base descriptor (§5).
memory layout	The mapping from guest address space to base-memory-object offsets (guest-address-linear); committed by the base descriptor (§5).
overlay-delta object	The canonical filesystem-delta encoding: content-object refs + namespace/metadata ops + whiteouts (§2.3, §6, §15).
startup-cohesion object	A per-image structural object that would pack hot startup-critical small-file bytes for minimal startup block count; RESERVED in v1 (packs are the v1 mechanism). If ever defined (LLCOH1): no dedup/warmth, additive, never the sole source of bytes (§4.4). Not a content object.
materialization strictness	Policy for how much state must be present before a lifecycle milestone, e.g. startup-only, readiness-critical, full-before-ready/exec, lazy-risk-accepted (§5, §7).
STARTUP_READY	The materialization milestone at which startup-critical metadata, proofs, and blocks are present so the workload can begin executing (§7).
attestation envelope	The runtime-provenance signature over a State Root identifier (signer, trust root, subject, validation; §2.5).
sandbox pin	Per-architecture, content-addressed (external sha256) identity of the runtime assets that produced a snapshot; a committed field constraining restore placement (§2.5).
assurance mode	The base's supply-chain assurance level (asserted / derived-attested / node-derived; §3, §14); a committed field of the base descriptor.
workload identity	The canonical digest of the restore-visible workload fields (pause image, hostname, namespace flags, pod/container security context, and per container name/image/command/args/env, working dir, run-as user, capabilities, mount/device topology, and resource limits; LLWI1 v2, §15.8); a committed field of a State Root (§6).
run mode	Which planes persist across a lifecycle event: clean, golden, persist-rootfs, or persist-rootfs+memory; a committed field of a State Root (§6.6). NB "golden" is a run-mode name (boot from a shared base memory snapshot), NOT an object class; "golden memory" as an object is retired.

producer	The identity recorded in a State Root as having produced it. The attestation-envelope signer MUST equal the producer or be policy-authorized to attest for it (§2.5).
base	Shorthand for a base descriptor and the shared, read-only state it binds (base rootfs + base memory snapshot).
node trust boundary	The set of components (kernel, runtime, imagefsd, trust roots) trusted on a node; the scope within which runtime attestation is issued and validated (§2.5, §14).
node score	The scheduler's per-node placement score combining cache-domain-scoped warmth minus node pressure (§10.2).
resume affinity	The scheduler preference for a node already holding an actor's local delta and/or warm base, for fast resume; advisory only (§10.3).
seed node	A node chosen to receive a prewarmed bootstrap/pack for a rollout or fork, from which others fan out (§10.4).
prewarm	Proactively fetching/verifying blocks (bootstrap, pack, base) onto nodes before they are needed (§7.4, §10.4).
mmap/exec-critical	Files/segments required for execve and dynamic linking (and selected mmap paths) that SHOULD be present before exec (§7, §10).
runtime adapter	A per-runtime layer (gVisor, Firecracker) that sources verified bytes from the CAS without changing internal identity or the verify-before-expose invariant (§11).
Sentry / gofer / lisafs	gVisor's user-space guest kernel (Sentry); the file-serving process it talks to (gofer) over the lisafs protocol; LLIFS serves the read-only lower through a CAS-backed gofer (§11.2).
MemoryFile	gVisor's guest-memory backing; the shared copy-on-write base capability (GVISOR-3) extends it to map a shared verified base file (§11.2).
platform mode	The gVisor execution platform (e.g. ptrace/systrap/KVM); a compatibility-matrix field (§11.4).
compatibility matrix	The base descriptor's record of what a restore requires (runtime+version, platform mode, arch+page size, CPU features, checkpoint format, ABI, seccomp/cgroup/ns); a mismatch fails closed (§11.4).
native backend	A later page-cache-integrated FS backend (EROFS/composefs/fs-verity/virtiofs) that preserves Terrapin identity and cache-domain policy (§11.5).
BlockRef	The canonical address of a block: object-digest / level / blockIndex (level 0 = data, 1+ = hash-file). The unit of coverage, packs, and fetch scheduling.
P0	The highest fetch priority class: a synchronous foreground read or page fault that the workload is blocked on; P0 preempts background hydration (§8).
restore working set	The access-ordered set of base memory snapshot blocks to prehydrate before resume; the memory-plane analog of the startup pack (§5.5, §7).
base version	An opaque identifier for one captured base (a base descriptor and its objects); base versions are few and each is stored as its own object (§5).
memory profile	The profile keying a restore working set to a base version (analogous to a startup profile, §7).
HAZARDS_CLEARED	The memory-restore lifecycle milestone at which RHAZARD-1..7 have been refreshed/invalidated so a restored agent may serve (§5.6, §5.7).

Retired terms (do not use): chunk, chunk map, artifact, plane manifest, opaque checkpoint, checkpoint plane, checkpoint manifest, golden memory (as an object), state-root-sha256 (as a separate scheme), "co-design"/team references for runtime changes (state required runtime capabilities technically, §11).